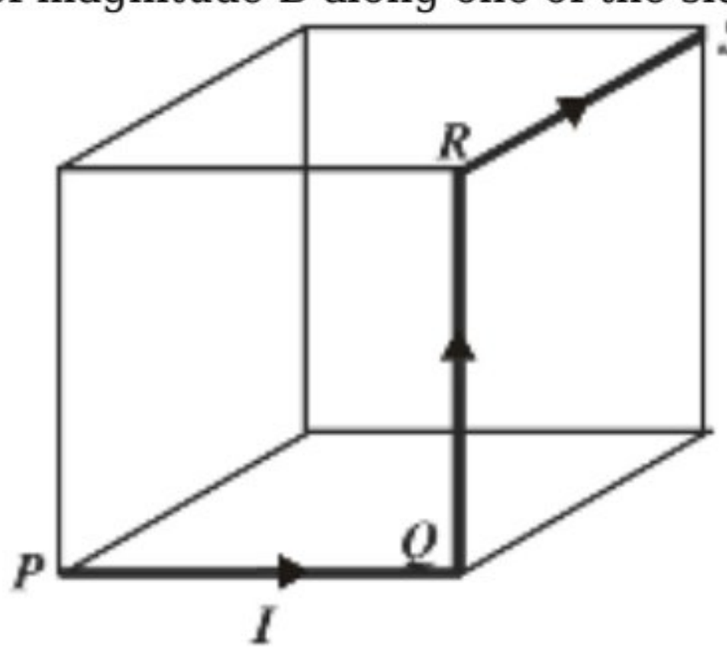
PHYSICS

1) A wire *PQRS* carrying a current *I* runs along three edges of a cube of side □ as shown. There exists a uniform magnetic field of magnitude B along one of the sides of the cube. The magnitude of



the force acting on the wire is

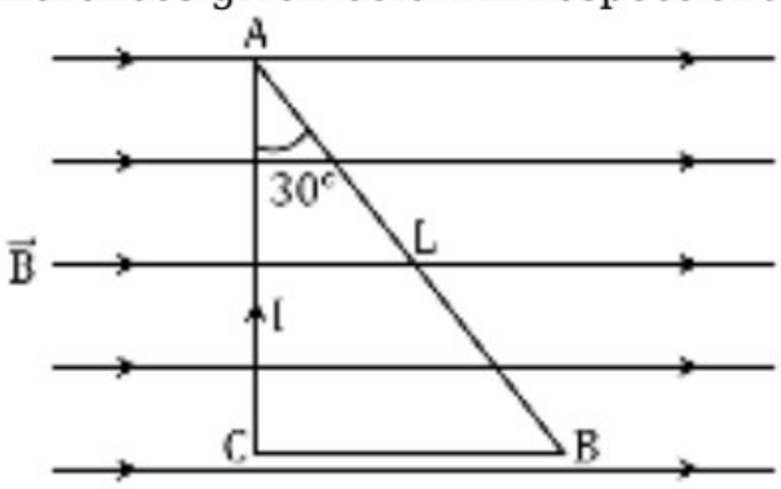
(1) 0

(2) I□B

(3) $\sqrt{2}I\square B$

(4) $2I\square B$

2) Consider the inferences given below in respect of the following current loop of wire kept in a magnetic field $\vec{B}$



(1) The magnitude and direction of force on the element AC of the wire is $\frac{\sqrt{3}}{2}$ILB directed into the page

(2) The magnitude and direction of force on the element AB of the wire is $\frac{\sqrt{3}}{2}$ILB directed into the page

(3) The total force on the loop ABCA is zero

Which of the above is/are not true

(1) 1 and 2

(2) 2 only

(3) 1 and 3

(4) 2 and 3

3) A particle of mass m and charge q moves with a constant velocity v along the positive x direction. It enters a region containing a uniform magnetic field B directed along the negative z direction,

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extending from x = a to x = b. What is the value of v required so that the particle can just enter the region x > b is

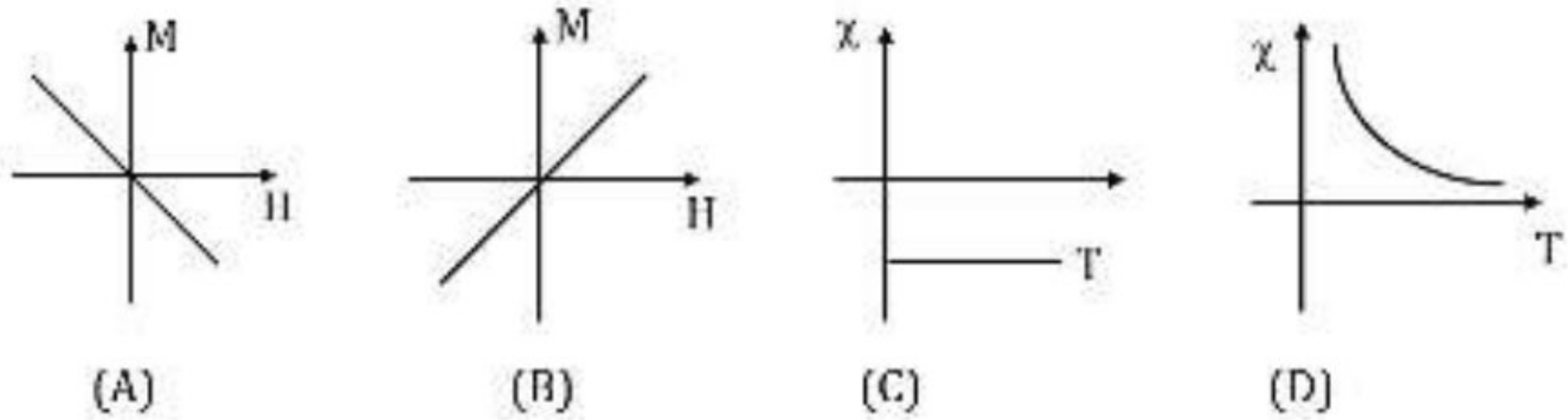
(1) $v > \frac{dqB}{m}$

(2) $v > \frac{q(b-a)B}{m}$

(3) $v > \frac{dqa}{m}$

(4) $v > \frac{q(b+a)B}{m}$

4) Following plots show Magnetisation (M) vs. Magnetising field (H) and Magnetic susceptibility ( ) vs Temperature (T) graphs.



(A) (B) (C) (D)

Which of the following combinations will be represented by a diamagnetic material ?

(1) (A), (D)

(2) (B), (C)

(3) (B), (D)

(4) (A), (C)

5) A charged particle going around in a circle can be considered to be a current loop. A particle of mass m carrying charge q is moving in a plane with speed v under the influence of magnetic field $\vec{B}$. The magnetic moment of this moving particle is

(1) $\frac{mv^2}{2}$

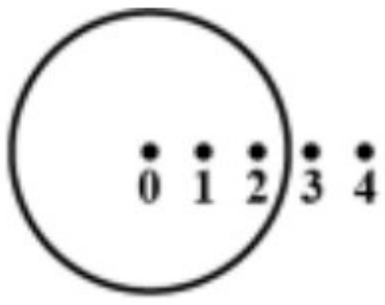
(2) $\frac{mvR}{2B}$

(3) $\frac{mv}{2}$

(4) $\frac{mv^2}{2B}$

6) The adjacent figure shows the cross-section of a long rod with its length perpendicular to the plane of the paper. It carries a constant current flowing along its length. B_1, B_2, B_3 and B_4 respectively represent the magnetic fields due to the current in the rod at point 1, 2, 3 and 4 lying at different separations from the centre O, as shown in the figure. Which of the following shall hold

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true

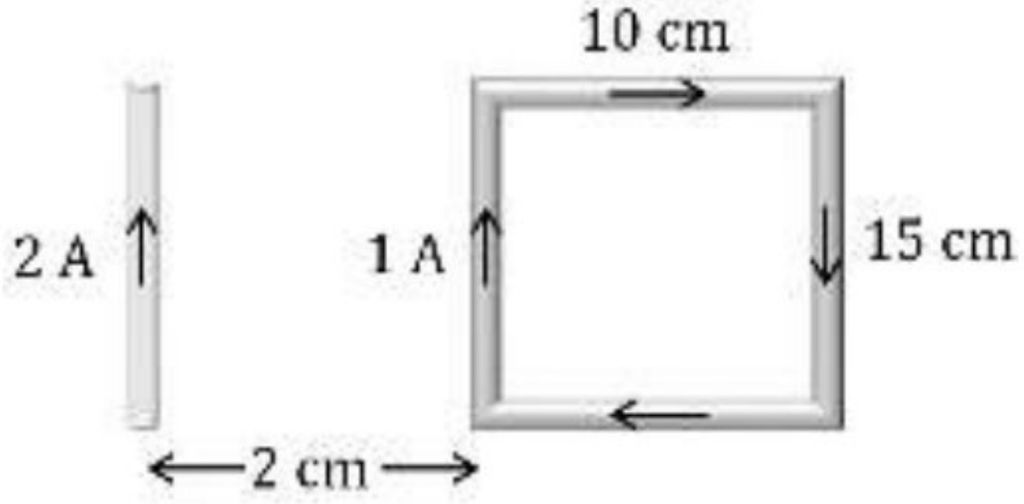
(1) $B_1 > B_2 \neq 0$

(2) $B_2 > B_3 \neq 0$

(3) $B_1 = B_2 = B_3 = 0$

(4) $B_3 > B_4 \neq 0$

7) What is the net force on the square coil



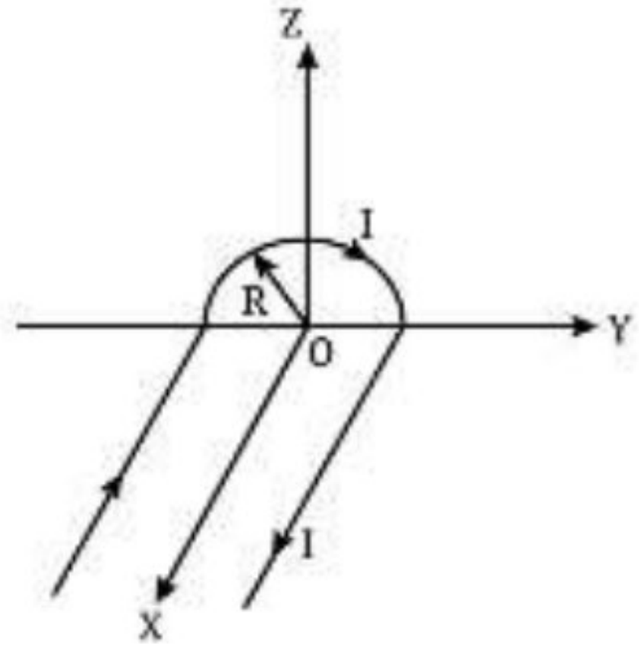
(1) 25×10^{-7} N moving towards wire

(2) 25×10^{-7} N moving away from wire

(3) 35×10^{-7} N moving towards wire

(4) 35×10^{-7} N moving away from wire

8) A wire carrying current I has the shape as shown adjoining figure. Linear parts of the wire are very long and parallel to X-axis while semicircular portion of radius R is lying Y-Z plane. Magnetic



field at point O is

(1) $\vec{B} = -\frac{\mu_0}{4\pi}\frac{I}{R}(\mu\hat{i} \times 2\hat{k})$

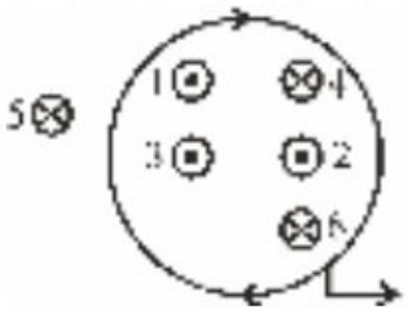
(2) $\vec{B} = -\frac{\mu_0}{4\pi}\frac{I}{R}(\pi\hat{i} + 2\hat{k})$

(3) $\vec{B} = \frac{\mu_0}{4\pi}\frac{I}{R}(\pi\hat{i} - 2\hat{k})$

(4) $\vec{B} = \frac{\mu_0}{4\pi}\frac{I}{R}(\pi\hat{i} + 2\hat{k})$

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9) The figure shown has a hypothetical amperian loop with the direction of movement marked. Six infinitely long wires having direction of current as shown and magnitude as follows are placed



perpendicular to the loop :-

$I_1 = 1A$ $I_4 = 1A$

$I_2 = 2A$ $I_5 = 5A$

$I_3 = 3A$ $I_6 = 4A$

The value of $\oint \vec{B}.\vec{dl}$ around the closed amperian loop will be :-

(1) μ_0 Wb/m

(2) $-\mu_0$ Wb/m

(3) $4\mu_0$ Wb/m

(4) $-4\mu_0$ Wb/m

10) A long solenoid has 200 turns per cm and carries a current i. The magnetic field at its centre is 6.28×10^{-2} weber/m^2. Another long solenoid has 100 turns per cm and it carries a current i/3. The value of the magnetic field at its centre is-

(1) 1.05×10^{-2} weber/m^2

(2) 1.05×10^{-5} weber/m^2

(3) 1.05×10^{-3} weber/m^2

(4) 1.05×10^{-4} weber/m^2

11) A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is :
($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1})

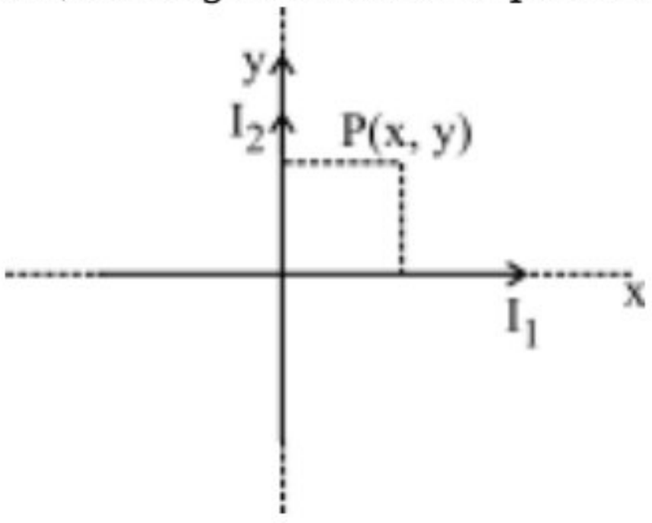
(1) 3.14×10^{-5} T

(2) 6.28×10^{-4} T

(3) 3.14×10^{-4} T

(4) 6.28×10^{-5} T

12) If magnetic field at point P is zero. Find relation between x and y of coordinate of point P.



(1) $y = \frac{I_1}{I_2} x$

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(2) $y = \frac{I_2}{I_1} x$

(3) $xy = I_1 I_2$

(4) None of these

13) A long straight wire of radius a carries a steady current I. The current is uniformly distributed over its cross-section. The ratio of the magnetic fields B and B', at radial distances $\frac{a}{2}$ and 2a respectively, from the axis of the wire is:

(1) $\frac{1}{4}$:1

(2) $\frac{1}{2}$:3

(3) 1:1

(4) 4:3

14) For a toroid of mean radius 20 cm, turns 240 and a current of 2A. What is the value of magnetic field within the toroid?

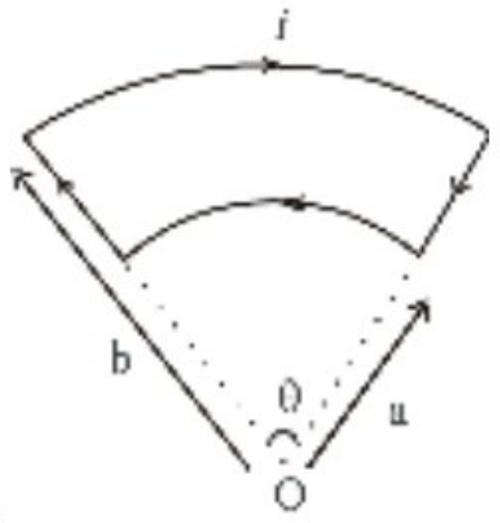
(1) 4.8×10^{-4} T

(2) 2.4×10^{-4} T

(3) 1.2×10^{-4} T

(4) Zero

15) Figure shows current loop having two circular arcs joined by two radial lines. Find magnetic field 'B' at center O.



(1) $\frac{\mu_0 i\theta}{4\pi}\left[\frac{b-a}{ab}\right]\odot$

(2) $\frac{\mu_0 i\theta}{4\pi}\left[\frac{b-a}{ab}\right]\otimes$

(3) $\frac{\mu_0 i\theta}{2\pi}\left[\frac{b-a}{ab}\right]\odot$

(4) $\frac{\mu_0 i\theta}{2\pi}\left[\frac{b-a}{ab}\right]\otimes$

16) A beam of ions with velocity 4×10^5 m/s enters normally into a uniform magnetic field of 8×10^{-2} tesla. If the specific charge (charge/mass) of the ion is 5×10^7 C/kg. Then the radius of circular path described will be :

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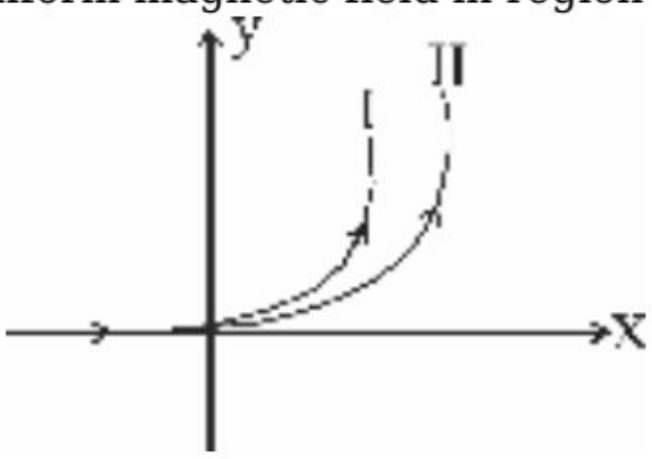
(1) 0.10 m

(2) 0.16 m

(3) 0.20 m

(4) 0.25 m

17) Three particles proton, α-particle and deutron are moving with same magnitude of momentum. There exist a uniform magnetic field in region x > 0. Figure shows path followed by the particles in



magnetic field :

(1) Proton will be along path I

(2) Deutron will be along path II

(3) α-particle will be along path II

(4) Magnetic field should be along $\hat{j}$

18) Currents of a 10 ampere and 2 ampere are passed through two parallel thin wires A and B respectively in opposite directions. Wire A is infinitely long and the length of the wire B is 2m. The force acting on the conductor B, which is situated at 10 cm distance from A will be :

(1) $8\pi \times 10^{-7}$ N

(2) 8×10^{-5} N

(3) $4\pi \times 10^{-7}$ N

(4) 5×10^{-5} N

19) An electron is moving along + x direction with a velocity of 6×10^{6} ms^{-1}. It enters a region of uniform electric field of 300 V/cm pointing along + y direction. The magnitude and direction of the magnetic field set up in this region such that the electron keeps moving along the x direction will be:

(1) 5×10^{-3} T, along +z direction

(2) 3×10^{-4} T, along –z direction

(3) 3×10^{-4} T, along +z direction

(4) 5×10^{-3} T, along –z direction

20)

A charged particle is projected in uniform magnetic field B with velocity v, having charge q perpendicular to the magnetic field. Then the force on charged particle is given by $F = q\left(\vec{v} \times \vec{B}\right)$.

Comment about the angular frequency of charged particle :-

(1) remians constant for every charged particle

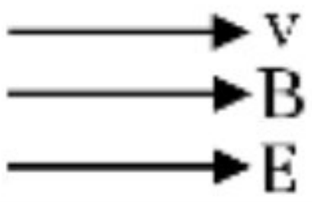
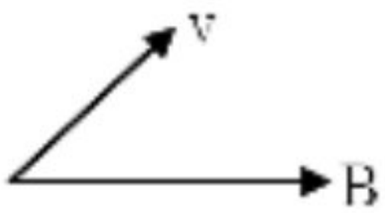
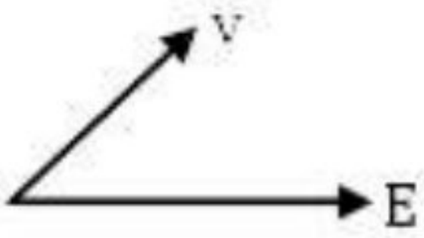
(2) increases with increase in radius of path

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(3) increase with decrease in radius of path

(4) remains constant whatever may be velocity of projection of given charged particle

21) **Column - I** represent a charge particle being thrown in electric field and magnetic field combination. **Column - II** represent the path followed by charge particle.

	Column - I		**Column - II**
(A)		(P)	Straight line
(B)		(Q)	Circle
(C)		(R)	Parabola
(D)		(S)	Helix

(1) (A) - (P); (B) - (Q); (C) - (S); (D) - (R)

(2) (A) - (P); (B) - (Q); (C) - (R); (D) - (S)

(3) (A) - (P); (B) - (R); (C) - (Q); (D) - (S)

(4) (A) - (P); (B) - (S); (C) - (R); (D) - (Q)

22) If an electron of velocity $(2\hat{i} + 3\hat{j})$ is subjected to magnetic field of $4\hat{k}$, then its-

(a) path will change

(b) speed does not change

(c) path must be circular

(d) momentum is constant

(1) a, b

(2) all

(3) a, b, c

(4) None

23) A charge particle having charge 2C is thrown with velocity of $(2\hat{i} + 3\hat{j})$ m/s inside a region having $\vec{E} = 2\hat{j}$ N/C and magnetic field $5\hat{k}$T. The Lorentz force acting on particle is

(1) $(30\hat{i} - 16\hat{j})$N

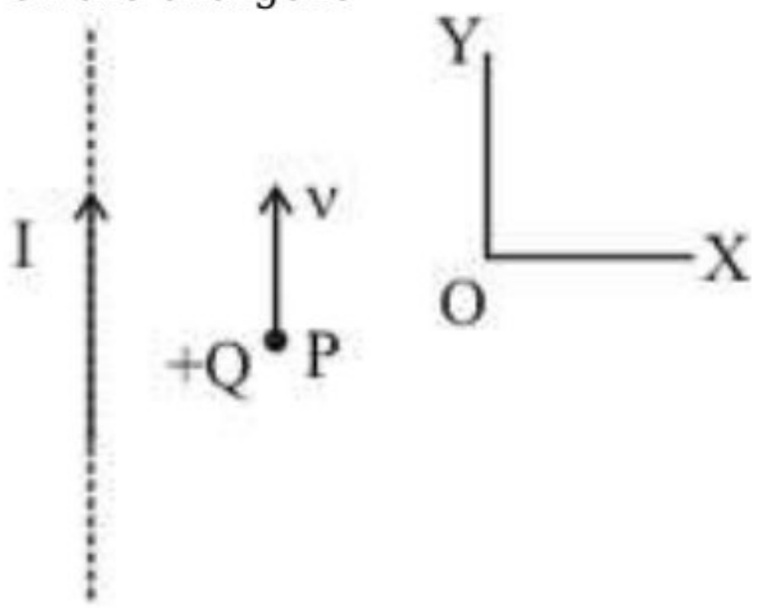
(2) $(15\hat{i} + 20\hat{j})$N

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(3) $(15\hat{i} - 30\hat{j})N$

(4) $(30\hat{i} + 15\hat{j})N$

24) A charge Q moves parallel to a very long straight wire carrying a current I as shown. The force on the charge is



(1) Opposite to OX

(2) Along OX

(3) Opposite to OY

(4) Along OY

25) A particle of charge q and mass m is projected in a uniform magnetic field $\vec{B} = 5\hat{i}T$ with velocity $\vec{v} = \left(2\hat{i} + 3\hat{j}\right)$ m/s. Find the pitch of the path followed.

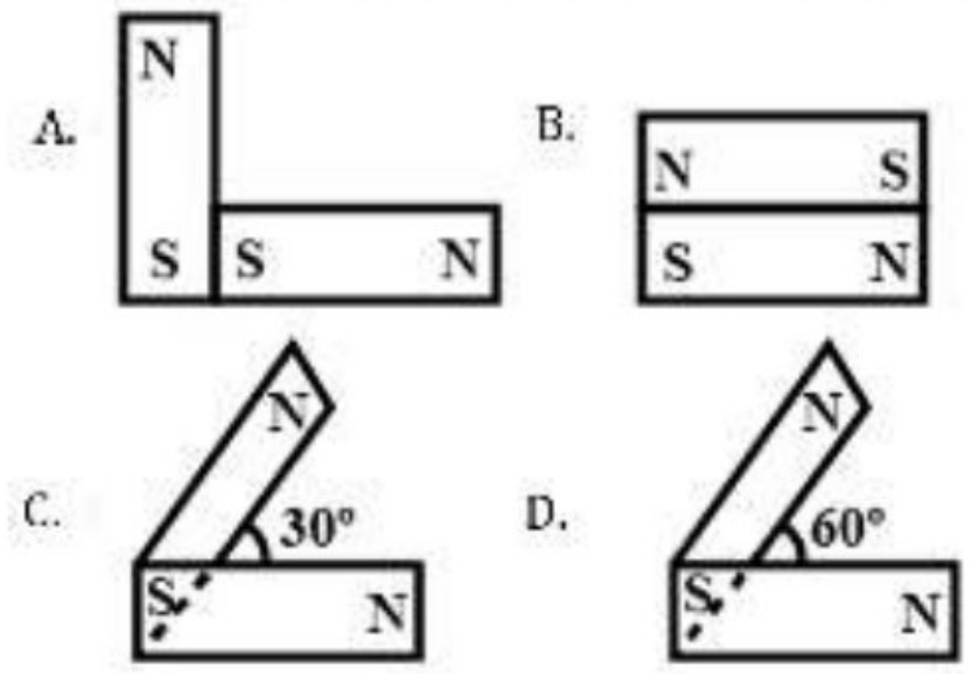
(1) $\frac{4\pi m}{5q}$

(2) $\frac{2\pi m}{5q}$

(3) $\frac{2\pi m}{3q}$

(4) $\frac{4\pi m}{2q}$

26) Following figures show the arrangement of bar magnets in different configuration. Each magnet has magnetic dipole moment $\vec{m}$. Which configuration has highest net magnetic dipole moment



(1) C

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(2) D

(3) A

(4) B

27) The magnetic field at a point X on the axis of a small bar magnet is equal to the field at a point Y on the equator of the same magnet. The ratio of the distances of point X and Y from the centre of the magnet is :-

(1) 2^{-3}

(2) $2^{-1/3}$

(3) 2^{3}

(4) $2^{1/3}$

28) A straight wire carrying current I is made into a circular loop If M is the magnetic moment associated with the loop, then the length of the wire is

(1) $\sqrt{\frac{4\pi M}{I}}$

(2) $\sqrt{\frac{2\pi M}{I}}$

(3) $\sqrt{\frac{\pi M}{I}}$

(4) $\sqrt{\frac{\pi M}{2I}}$

29) A magnet is parallel to a uniform magnetic field. If it is rotated by 60°, the work done is 0.8 J. How much work is done in moving it 30° further

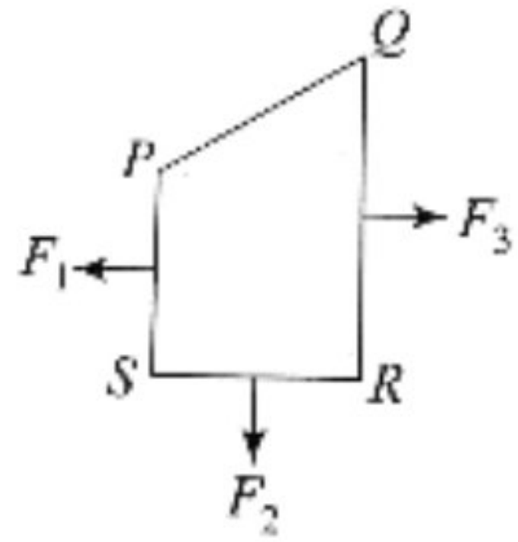
(1) 0.8×10^{7} ergs

(2) 0.4 J

(3) 8 J

(4) 0.8 ergs

30) A closed loop PQRS carrying a current is placed in a uniform magnetic field. If the magnetic forces on segment PS, SR and RQ are F_1, F_2 and F_3 respectively and area in the plane of the paper



and along the directions shown, the force on the segment QP is

(1) $\sqrt{(F_3 - F_1)^2 - F_2^2}$

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(2) $F_3 + F_1 - F_2$

(3) $F_3 - F_1 + F_2$

(4) $\sqrt{(F_3 - F_1)^2 + F_2^2}$

31) A flywheel rotates with a uniform angular acceleration. Its angular velocity increases from 20π rad/s to 40π rad/s in 10 sec. How many rotations did it make in this period :-

(1) 50

(2) 150

(3) 100

(4) 250

32) When a particle moves in a circle with uniform speed :-

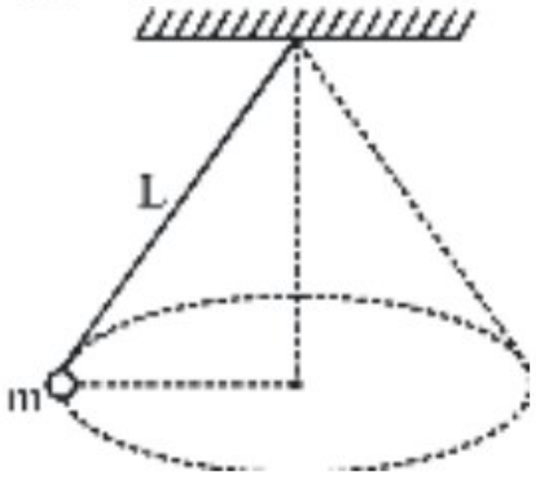
(1) its velocity and acceleration are both constant

(2) its velocity is constant but the acceleration changes

(3) its acceleration is constant but the velocity changes

(4) its velocity and acceleration both change

33) A ball of mass (m) 0.5 kg is attached to the end of a string having length (L) 0.5 m. The ball is rotated on a horizontal circular path about vertical axis. The maximum tension that the string can bear is 324 N. The maximum possible value of angular velocity of ball (in radian/s) is :-



(1) 9

(2) 18

(3) 27

(4) 36

34)

A stone is moved round a horizontal circle with a 20 cm long string tied to it. If centripetal acceleration is 9.8 m/s^2, then its angular velocity will be-

(1) 7 rad/s

(2) 22/7 rad/s

(3) 49 rad/s

(4) 14 rad/s

35) A Rotor of radius 2 m rotates such that a person stays pinned to the wall. If g decreases by

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10% on a different planet, by what percentage must the minimum angular velocity ω change?

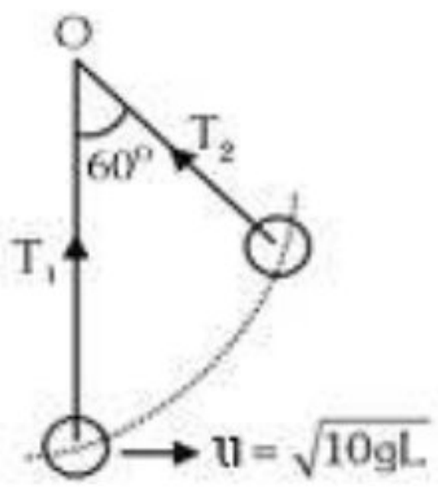
(1) Increase by 10%

(2) Decrease by 10%

(3) Decrease by 5.1%

(4) Increase by 4.9%

36) A ball of mass m is attached to a string of length L and given a horizontal velocity $\sqrt{10gL}$ at its lowest point as shown. The ratio of tension in the string when it has turned an angle 60° to initial tension, i.e. T_2/T_1 is :



(1) $\frac{17}{21}$

(2) $\frac{19}{21}$

(3) $\frac{19}{22}$

(4) $\frac{17}{23}$

37) A stone is tied at one end of a 5 m long string and whirled in a vertical circle. The minimum speed at the top required to just complete the vertical circle is : ($g = 9.8\ m/s^2$)

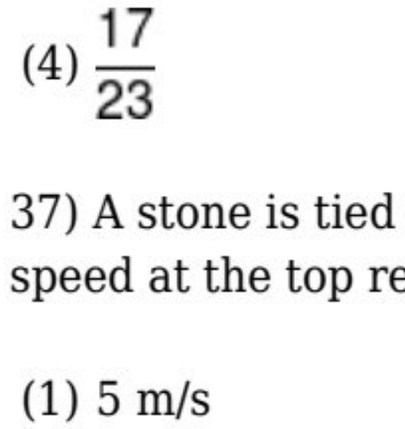
(1) 5 m/s

(2) $5\sqrt{7}$ m/s

(3) 7 m/s

(4) $7\sqrt{5}$ m/s

38) A bob of simple pendulum of length □ is moving in a vertical circle about point O such that tension in string at top most point B is zero. The acceleration of bob at point A is ($g = 10\ m/s^2$) :



(1) $10\sqrt{5}\ m/sec^2$

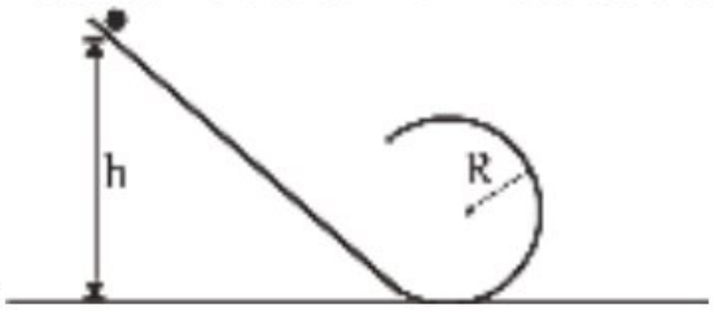
(2) $10\sqrt{10}\ m/sec^2$

(3) $10\ m/sec^2$

(4) $5\sqrt{10}\ m/sec^2$

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39) In the figure is a smooth track which has a circular part of radius of curvature 0.1 m. The minimum height from where a small ball must be released from rest so that it completes the circle without leaving contact with the track, is



(1) 0.1 m

(2) 0.2 m

(3) 0.25 m

(4) 0.4 m

40) Two particles A and B are projected from ground with speed 2 m/s and $\frac{4}{\sqrt{3}}$ m/s from horizontal with angle 30° and 60° respectively. If seperation between A and B on ground is 10 m. Find relative angular velocity of B w.r.t A :

(1) 0.1 rad/s

(2) 0.2 rad/s

(3) 0.05 rad/s

(4) 1 rad/s

41) What is the value of linear velocity, if $\vec{\omega} = 3\hat{i} - 4\hat{j} + \hat{k}$ and $\vec{r} = 5\hat{i} - 6\hat{j} + 6\hat{k}$:-

(1) $6\hat{i} + 2\hat{j} - 3\hat{k}$

(2) $-18\hat{i} - 13\hat{j} + 2\hat{k}$

(3) $4\hat{i} - 13\hat{j} + 6\hat{k}$

(4) $6\hat{i} - 2\hat{j} + 8\hat{k}$

42) Keeping the angle of banking same, if the radius of curvature is made four times, the percentage increase in the maximum speed with which a vehicle can travel on a circular road is :-

(1) 25%

(2) 50%

(3) 75%

(4) 100%

43) Three identical particles are joined together by a thread as shown in figure. All the three particles are moving in a horizontal plane. If the velocity of the outermost particle is υ_0, then the ratio of tensions in the three sections of the string is ($T_{BC} : T_{AB} : T_{OA}$)

(1) 3 : 5 : 7

(2) 3 : 4 : 5

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(3) 7 : 11 : 6

(4) 3 : 5 : 6

44) The breaking tension of a string is 10 N. A particle of mass 0.1 kg tied to it is rotated along a horizontal circle of radius 0.5 metre. The maximum speed with which the particle can be rotated without breaking the string is-

(1) $\sqrt{5}$ m/sec

(2) $\sqrt{(50)}$ m/sec

(3) $\sqrt{(500)}$ m/sec

(4) $\sqrt{(1000)}$ m/sec

45) A particle is kept at rest at top of a sphere of diameter 30 m. When disturbed slightly it slides down. At what height h from bottom, the particle will leave the sphere :-

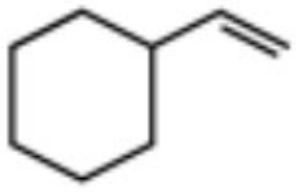
(1) 25 m

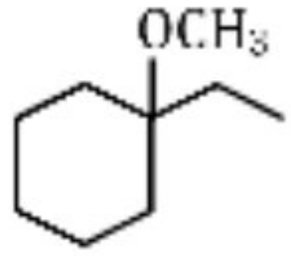
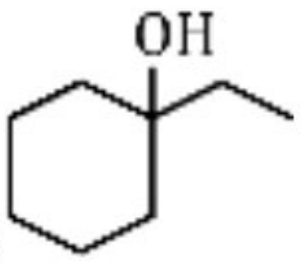
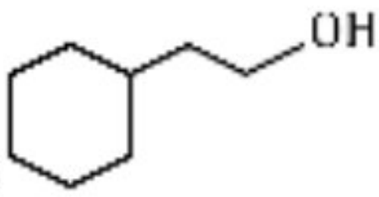
(2) 30 m

(3) 20 m

(4) 15 m

CHEMISTRY

1) A compound  is converted in compound listed in Column-I. The suitable reagent is listed in column-II.

Column-I (Product)	**Column-II (Reagent)**
(A) 	(P) $H^{\oplus}/H_2O$
(B) 	(Q) (i) $H\text{-}Cl/R_2O_2$ (ii) H_2O
(C) 	(R) (i) BH_3/THF (ii) $H_2O_2/\overset{\ominus}{O}H$
	(S) (i) $H^{\oplus}$ (ii) CH_3OH

Correct Match is

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(1) A → S ; B → P, Q ; C → R

(2) A → Q ; B → P ; C → R, S

(3) A → S ; B → P ; C → Q, R

(4) A → Q ; B → P ; C → R

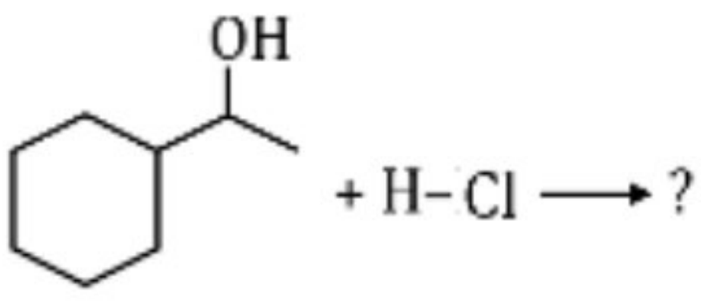
2) **Statement-I :** Tertiary alcohol produce turbidity immediately with Lucas reagent.
Statement-II : Primary alcohols do not produce turbidity at room temperature .

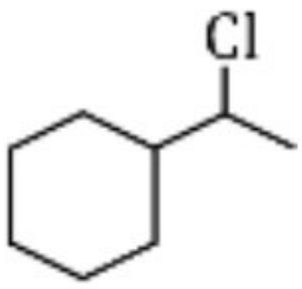
(1) Both Statement I and Statement II are true.

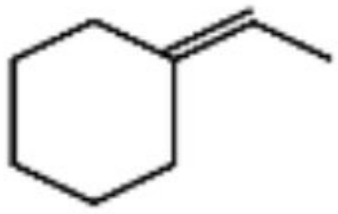
(2) Both Statement I and Statement II are false.

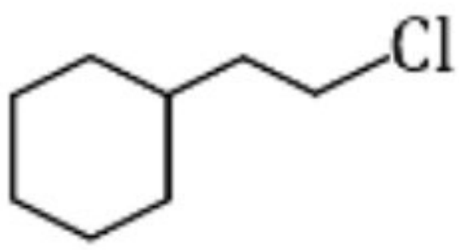
(3) Statement I is true but Statement II is false.

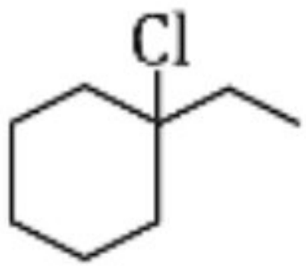
(4) Statement I is false but Statement II is true.

3) 

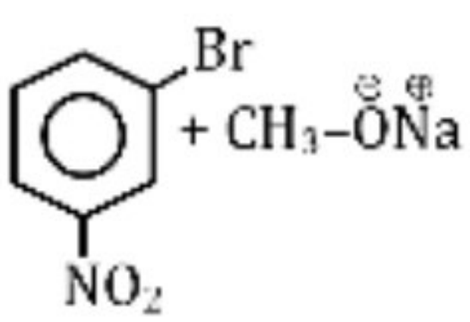
(1) 

(2) 

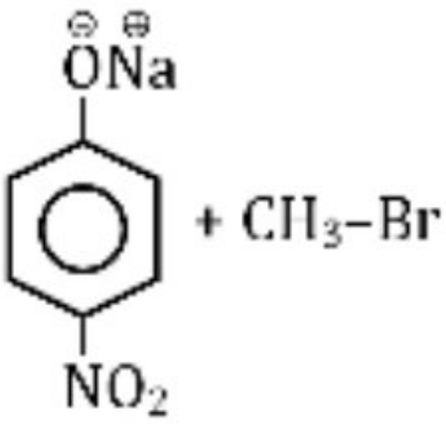
(3) 

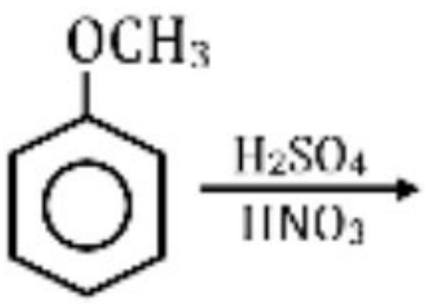
(4) 

4) Which is appropriate set of reactants for the preparation of 1-Methoxy-4-nitrobenzene

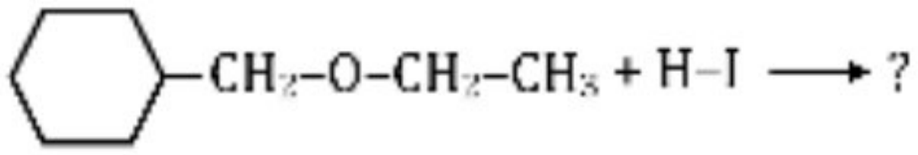
(1) 

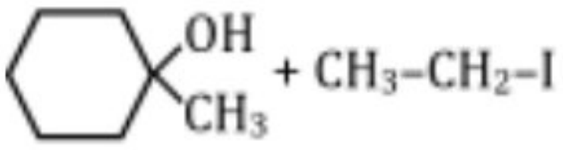
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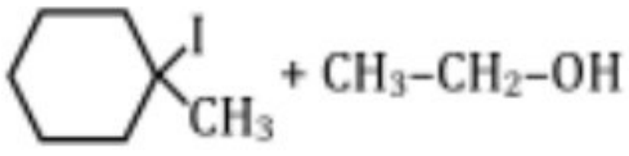
(2) 

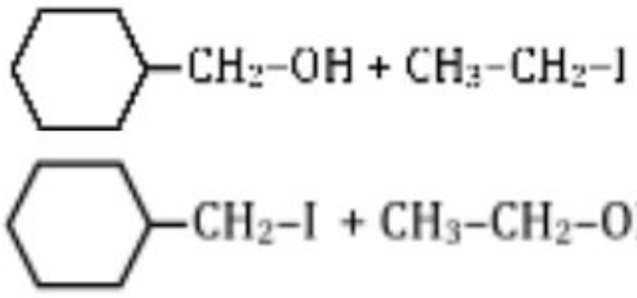
(3) 

(4) (2) and (3) Both

5) 

(1) 

(2) 

(3) 

(4)

6) Among the following sets of reactants which one produces anisole?

(1) CH_3CHO ; RMgX

(2) C_6H_5OH ; NaOH ; CH_3I

(3) C_6H_5OH ; neutral $FeCl_3$

(4) C_6H_5 – CH_3 ; CH_3COCl; $AlCl_3$

7) Reaction of phenol with chloroform in presence of dilute sodium hydroxide finally introduces which one of the following functional group ?

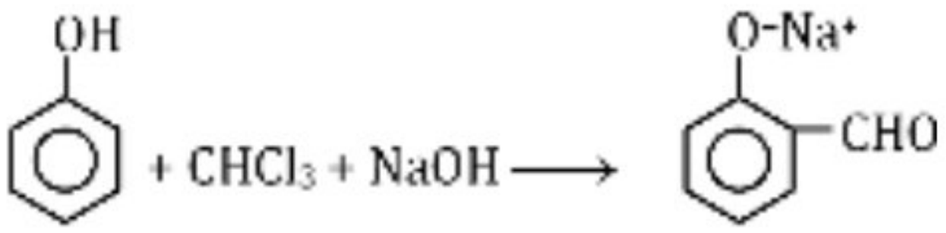
(1) $-CHCl_2$

(2) –CHO

(3) $-CH_2Cl$

(4) –COOH

8) In the reaction



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the electrophile involved is

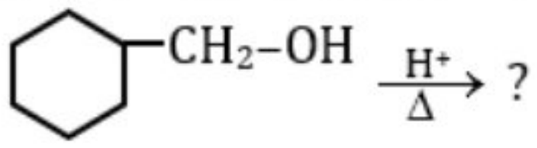
(1) Dichloromethyl cation $(\overset{\oplus}{C}HCl_2)$

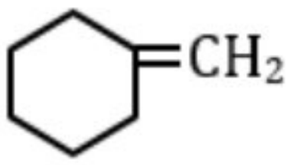
(2) Formyl cation $(\overset{\oplus}{C}HO)$

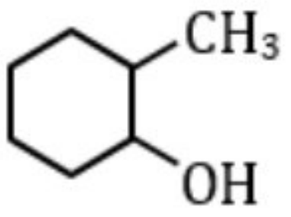
(3) Dichloromethyl anion $(\overset{\ominus}{C}HCl_2)$

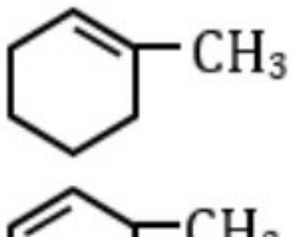
(4) Dichlorocarbene ($:CCl_2$)

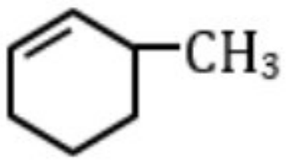
9) The major product in the following reaction is



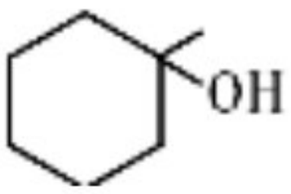
(1) 

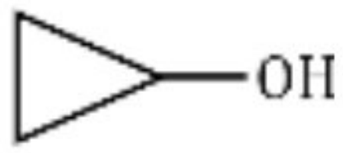
(2) 

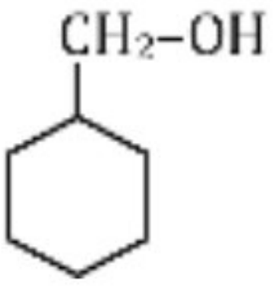
(3) 

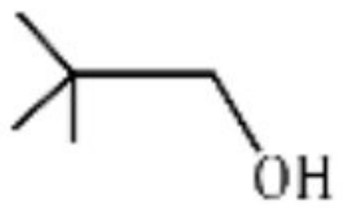
(4) 

10) Identify which of the following compound is 2° alcohol ?

(1) 

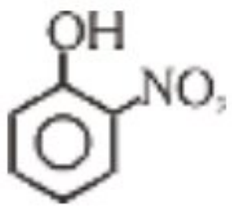
(2) 

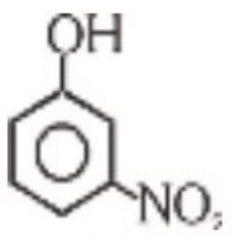
(3) 

(4) 

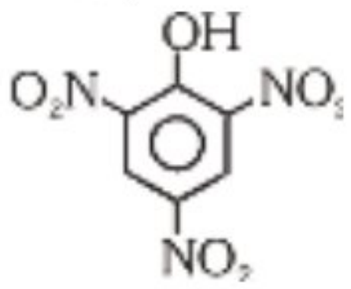
11) Nitration of phenol with conc. nitric acid gives :-

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(1) 

(2) 

(3) 

(4) 

12) CH_3–CH_2–OH + Ph–CH_2–OH $\xrightarrow[140°C]{H^{\oplus}}$ of which is not obtained?

(1) CH_3–CH_2–OCH_2–CH_3

(2) Ph–CH_2–OCH_2–Ph

(3) Ph–CH_2–O–CH_2–CH_3

(4) Ph–CH_2–O–CH_2–O–CH_3

13) Phenol and benzoic acid are distinguished by :-

(1) Lucas reagent

(2) Victor Meyer test

(3) Hinsberg reagent

(4) Sodium bicarbonate

14) The reaction



is called :-

(1) Schotten Bauman reaction

(2) Kolbe Schmidt reaction

(3) Reimer–Tiemann reaction

(4) Lederer–Manasse reaction

15) Which of the following is insoluble in water ?

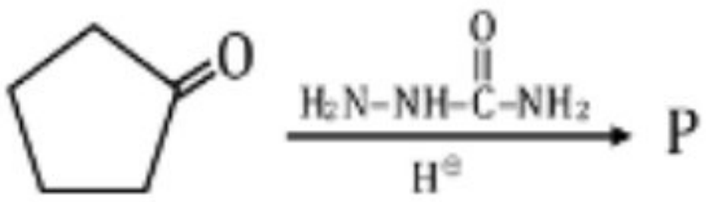
(1) Ethanol

(2) Ethoxyethane

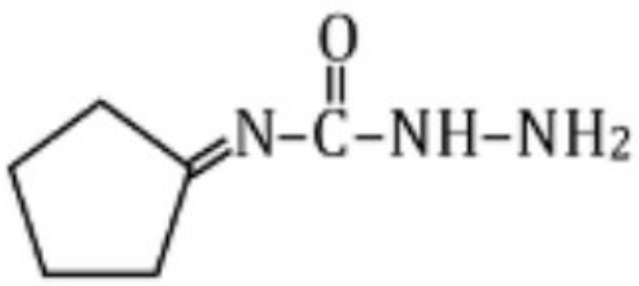
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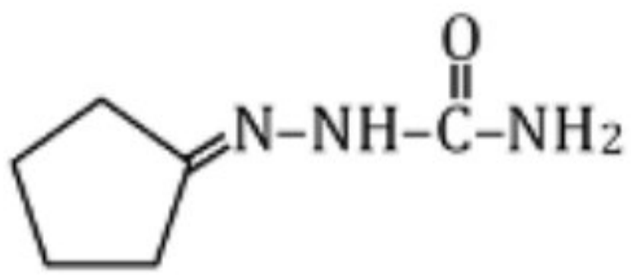
(3) Phenol

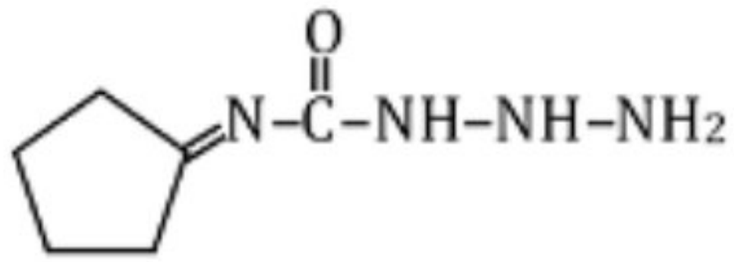
(4) Pentane

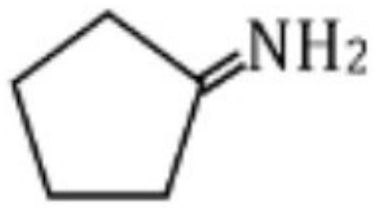
16) 

what is "P" ?

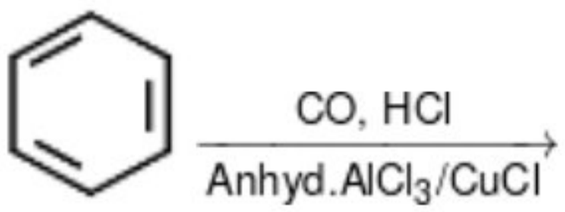
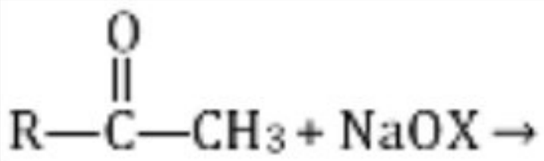
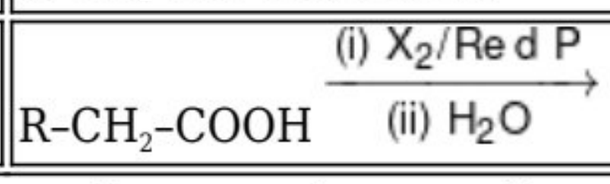
(1) 

(2) 

(3) 

(4) 

17) Match List-I with List-II.

List-I		List-II	
(a)		(i)	Hell-Volhard-Zelilnskyreaction
(b)		(ii)	Gattermann-Koch Reaction
(c)		(iii)	Haloform reaction
(d)		(iv)	Esterification

Choose the correct answer from the options given below.

(1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

(4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

18) Which of the following is most reactive towards S_NAE reaction ?

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(1) Acid halide

(2) Amide

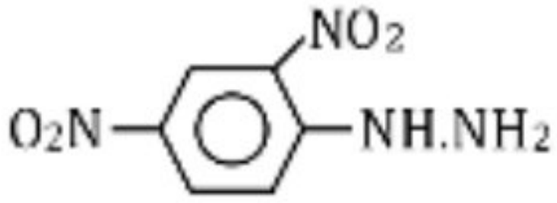
(3) Anhydride

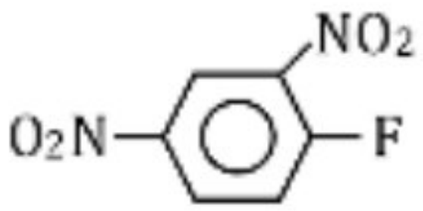
(4) Ester

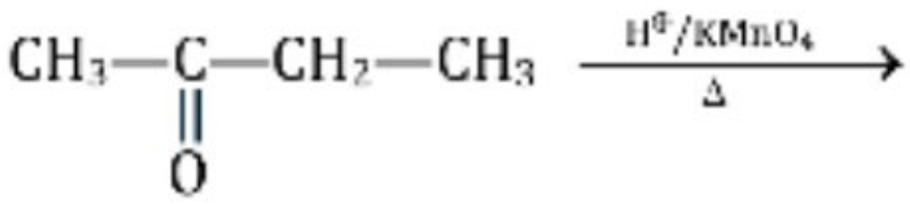
19) Brady's reagent is

(1) $[Cu(NH_3)_4]SO_4$

(2) $KMnO_4/NaIO_4$

(3) 

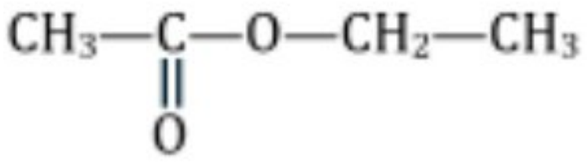
(4) 

20)  Product is -

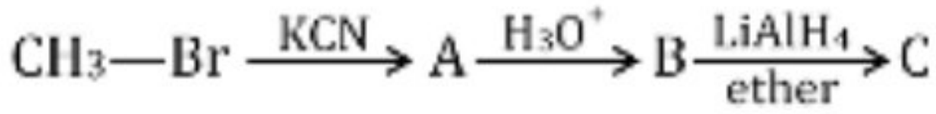
(1) Only CH_3—COOH

(2) CH_3—CH_2—COOH + CO_2

(3) H—COOH + CH_3—CH_2—COOH

(4) 

21) In the following sequence of reactions



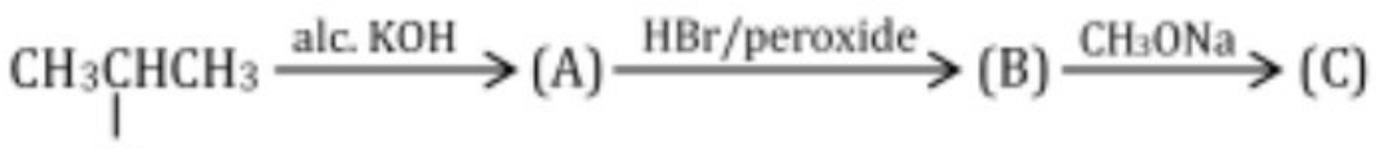
the end product (C) is :

(1) Acetaldehyde

(2) Ethyl alcohol

(3) Acetone

(4) Methane



22)

In the above reaction sequence, the product [C] is:

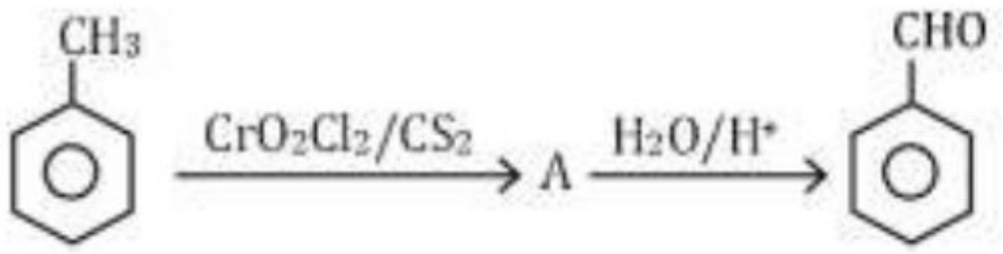
(1) Diethyl ether

(2) 1-methoxypropane

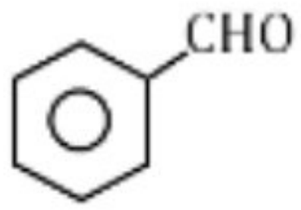
(3) Isopropyl alcohol

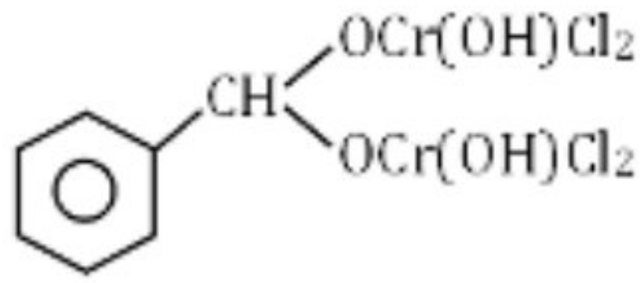
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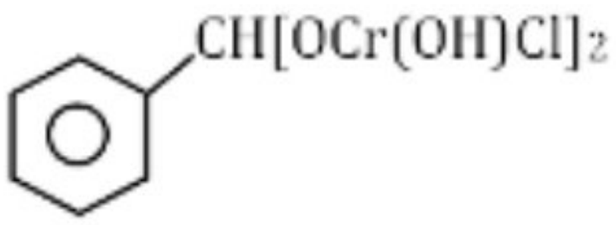
(4) Propylene glycol

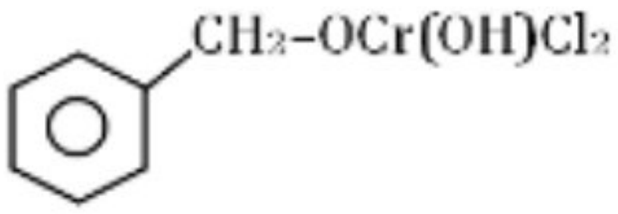
23) 

Intermediate product 'A' is-

(1) 

(2) 

(3) 

(4) 

24) When pentanone-2 reacts with ethyl magnesium chloride followed by hydrolysis it gives:-

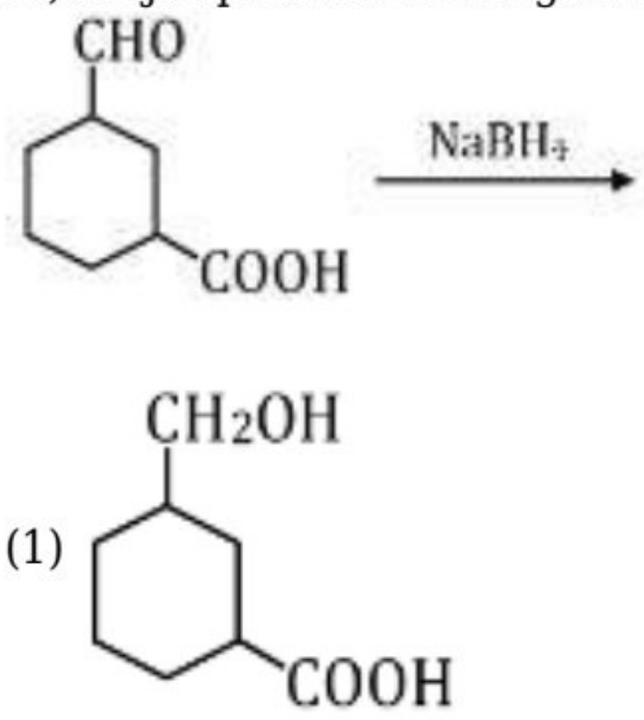
(1) Aldehyde

(2) Carboxylic acid

(3) Ketone

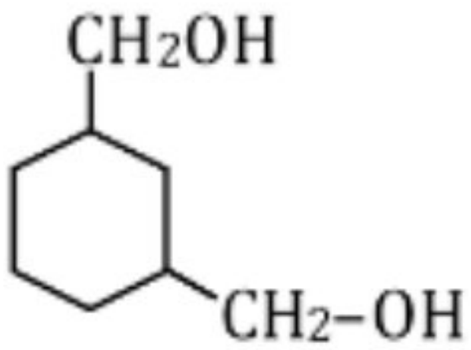
(4) alcohol

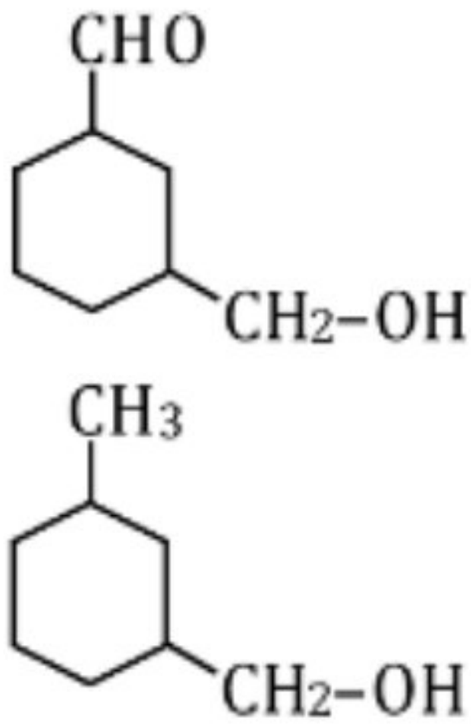
25) Major product of the given reaction is :



(1)

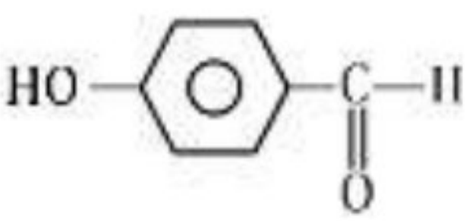
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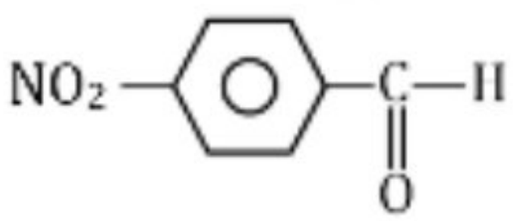
(2) 

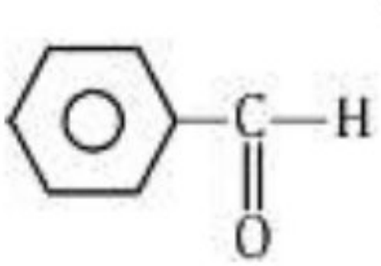
(3) 

(4)

26) Which of the following compounds is least reactive in nucleophilic addition reaction

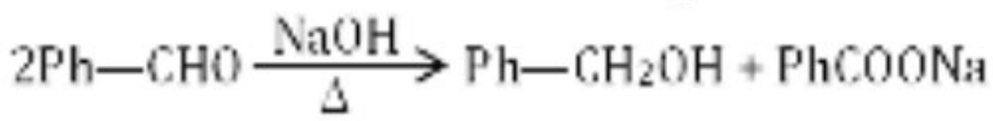
(1) 

(2) 

(3) 

(4) All are equally reactive

27) In the Cannizzaro reaction given below :



the slowest step is :-

(1) The attack of at the carbonyl group

(2) The transfer of hydride ion to the carbonyl group

(3) The abstraction of proton from the carboxylic acid

(4) The deprotonation of $Ph–CH_2OH$

28) Given below are two statements

Statement – I : 2,4-DNP-derivatives are yellow, orange or Red solids, useful for characterisation of aldehydes and ketones.

Statement – II : Reaction of carbonyl compound with 2,4-DNP reagent is an example of electrophilic addition reaction.

In the light of the above statements, choose the most appropriate answer from the options given

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below

(1) Both Statement I & II are correct

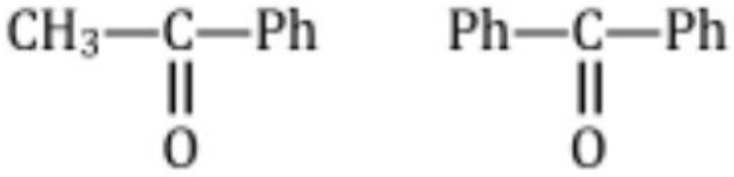
(2) Both Statement I & II are incorrect

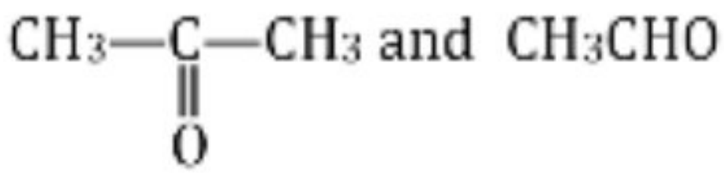
(3) Statement I is correct but Statement II is incorrect

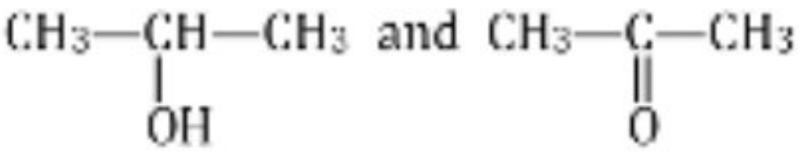
(4) Statement I is incorrect but Statement II is correct

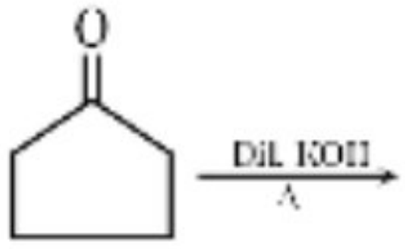
29) Which of the following pairs are distinguished by Iodoform test :-

(1) CH_3OH and $CH_3—CH_2—CH_2—OH$

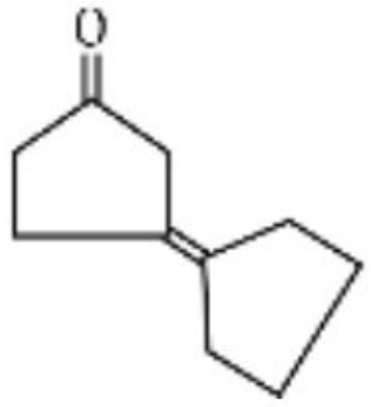
(2) 

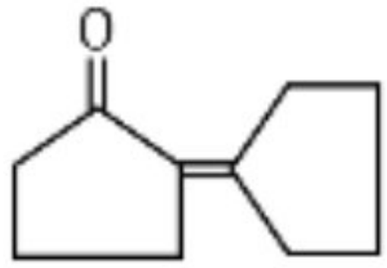
(3) 

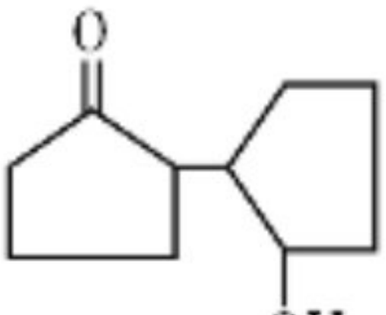
(4) 

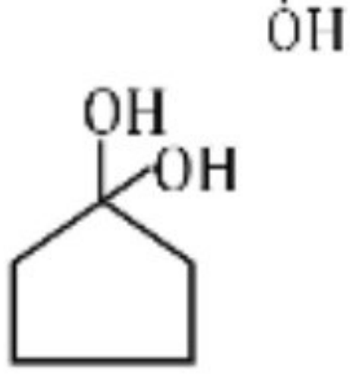
30) 

(A); Product (A) is :-

(1) 

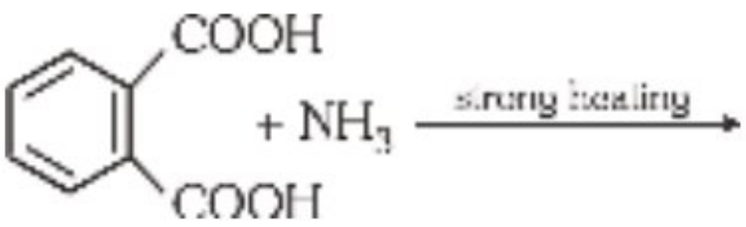
(2) 

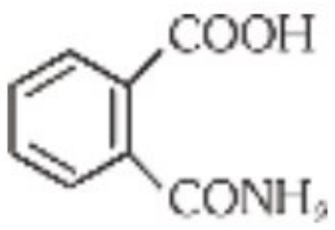
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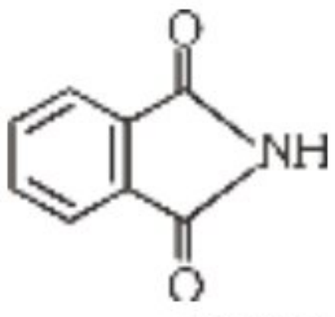
(4) 

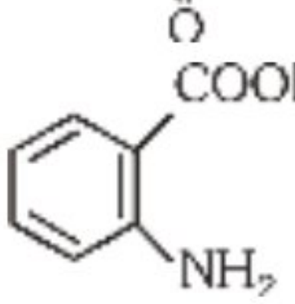
31) The major product of the following reaction is :

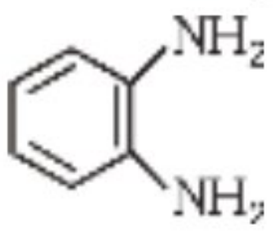
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(1) 

(2) 

(3) 

(4) 

32) On heating a mixture of common salt and $K_2 Cr_2 O_7$ in equal amount along with concentrated $H_2 SO_4$ in a test tube, a gas is evolved. Formula of the gas evolved and oxidation state of the central metal atom in the gas respectively are:

(1) $CrO_2 Cl_2$ and +5

(2) $CrO_2 Cl_2$ and +6

(3) $Cr_2 O_2 Cl_2$ and +6

(4) $Cr_2 O_2 Cl_2$ and +3

33) The set having ions which are coloured and paramagnetic both is:

(1) Cu^+ , Zn^{2+}, Mn^{4+}

(2) Sc^{3+}, V^{5+}, Ti^{4+}

(3) Ni^{2+} , Mn^{7+}, Hg^{2+}

(4) Cu^{2+}, Cr^{3+}, Sc^+

34) Observe the following statements:

a. The almost identical radii of Zr and Hf is consequence of Lanthanide contraction

b. Actinium is also a member of actinides

c. Copper (I) compounds are colourless except when colour results from charge transfer phenomenon

d. Eu^{2+} acts as a strong reducing agent

The correct statement is are

(1) only b

(2) only a, c and d

(3) only a and d

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(4) only a and b

35) Which of the following lanthanoid ions is diamagnetic? (at nos. Ce = 58, Sm = 62, Eu = 63, Yb = 70)

(1) Eu^{2+}

(2) Yb^{2+}

(3) Ce^{2+}

(4) Sm^{2+}

36) Which of the following electronic configuration would be associated with the highest magnetic moment ?

(1) [Ar] $3d^7$

(2) [Ar] $3d^8$

(3) [Ar] $3d^3$

(4) [Ar] $3d^6$

37) MnO_4^{1-} is of intense pink colour, through Mn is in +VII oxidation state. This is due to

(1) d-d transition

(2) charge transfer when Mn gives electron to oxygen

(3) charge transfer when oxygen gives its electrons to Mn making it +VI

(4) absorption of IR frequencies

38) Match List-I with List-II and select the correct answer using codes given below the lists:

List-I	**List-II**
Metal ions	Magnetic moment (B.M.)
(1) Cr^{3+}	i. $\sqrt{35}$
(2) Fe^{2+}	ii. $\sqrt{30}$
(3) Ni^{2+}	iii. $\sqrt{24}$
(4) Mn^{2+}	iv. $\sqrt{15}$
	v. $\sqrt{8}$

(1) (1) – i, (2) – iii, (3) – v, (4) – iv

(2) (1) – ii, (2) – iii, (3) – v, (4) – i

(3) (1) – iv, (2) – iii, (3) – v, (4) – i

(4) (1) – iv, (2) – v, (3) – iii, (4) – i

39) The outer electron configuration of Gd (At. no. 64) is:

(1) $4f^3, 5d^5, 6s^2$

(2) $4f^8, 5d^0, 6s^2$

(3) $4f^4, 5d^4, 6s^2$

(4) $4f^7, 5d^1, 6s^2$

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40) Which of the following group of transition metals is called coinage metals :-

(1) Cu, Ag, Au

(2) Ru, Rh, Pd

(3) Fe, Co, Ni

(4) Os, Ir, Pt

41) The magnetic moment of a transition metal compound has been calculated to be 3.87 B.M. The metal ion is

(1) Cr^{2+}

(2) Mn^{2+}

(3) V^{2+}

(4) Ti^{2+}

42) Which one of the following exhibits the largest number of oxidation states?

(1) Mn(25)

(2) V(23)

(3) Cr (24)

(4) Ti (22)

43) $(NH_4)_2\ Cr_2O_7$ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is :-

(1) CrO_2

(2) Cr_2O_3

(3) Cr_2O_4

(4) CrO_3

44) Which of the following set of elements have almost same radii :-

(1) Mo & W

(2) Ti & Zr

(3) Y & La

(4) V & Nb

45) Which of the following can also shows +4 oxidation state :-

(1) Ce & yb

(2) Eu & Yb

(3) Tb & Eu

(4) Ce & Tb

BIOLOGY

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1) **Statement I:** In monocotyledonous plants, the primary root is short-lived and is replaced by a large number of roots originating from the base of the stem.
Statement II: Adventitious roots are defined as roots that originate directly from the direct elongation of the radicle.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct, but Statement II is incorrect

(4) Statement I is incorrect, but Statement II is correct

2) Read the following statements regarding leaves and select the correct option:
Statement I: In a pinnately compound leaf, leaflets are attached at a common point at the tip of the petiole.
Statement II: A bud is present in the axil of the petiole of both simple and compound leaves, but not in the axil of leaflets.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct, but Statement II is incorrect

(4) Statement I is incorrect, but Statement II is correct

3) Read the following statements regarding the seed structure and select the correct option:
Statement I: In maize seeds, the outer covering of the endosperm separates the embryo by a proteinaceous layer called the aleurone layer.
Statement II: All monocotyledonous seeds are strictly endospermic without any exceptions.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct, but Statement II is incorrect

(4) Statement I is incorrect, but Statement II is correct

4) **Assertion (A):** In cymose inflorescence, the growth of the main axis is limited.
Reason (R): In cymose inflorescence, the main axis terminates in a flower.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

5) **Assertion (A):** Ray florets of sunflower have an inferior ovary.
Reason (R): In epigynous flowers, the margin of the thalamus grows upward enclosing the ovary completely and getting fused with it.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

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6) **Assertion (A):** A staminode is capable of producing fertile pollen grains in large quantities.
Reason (R): Stamens attached to petals are called epipetalous stamens.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

7) Which of the following plants exhibit alternate phyllotaxy?
1. China rose
2. *Calotropis*
3. Mustard
4. Sunflower
5. Guava

(1) 1, 3, and 4 only

(2) 1, 2, and 5 only

(3) 2, 3, and 4 only

(4) 1, 4, and 5 only

8)

Match Column-I (Type of Placentation) with Column-II (Example Plant):

Column-I	**Column-II**
a. Marginal	**(i)** *Dianthus*
b. Axile	**(ii)** Sunflower
c. Parietal	**(iii)** Pea
d. Free central	**(iv)** Lemon
e. Basal	**(v)** Mustard

(1) a-(iii), b-(v), c-(iv), d-(i), e-(ii)

(2) a-(iii), b-(iv), c-(v), d-(i), e-(ii)

(3) a-(v), b-(iv), c-(iii), d-(ii), e-(i)

(4) a-(iii), b-(iv), c-(i), d-(v), e-(ii)

9) Thread-like epidermal extensions that absorb water and minerals from the soil arise from which region of the root?

(1) Region of meristematic activity

(2) Root cap

(3) Region of elongation

(4) Region of maturation

10) Swollen leaf base found in some leguminous plants is termed as:

(1) Pulvinus

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(2) Sheath
(3) Stipule
(4) Petiole

11) The arrangement of veins and veinlets in the lamina of a leaf is known as:

(1) Phyllotaxy
(2) Aestivation
(3) Venation
(4) Placentation

12) Which is not a stem modification -

(1) Rhizome of ginger
(2) Corm of Colocasia
(3) Pitcher of *Nepenthes*
(4) Tuber of potato

13) In racemose inflorescence, flowers are borne in which order/succession?

(1) Basipetal order
(2) Centripetal order
(3) Acropetal succession
(4) Irregular arrangement

14) Which of the following flowers is zygomorphic (bilaterally symmetrical)?

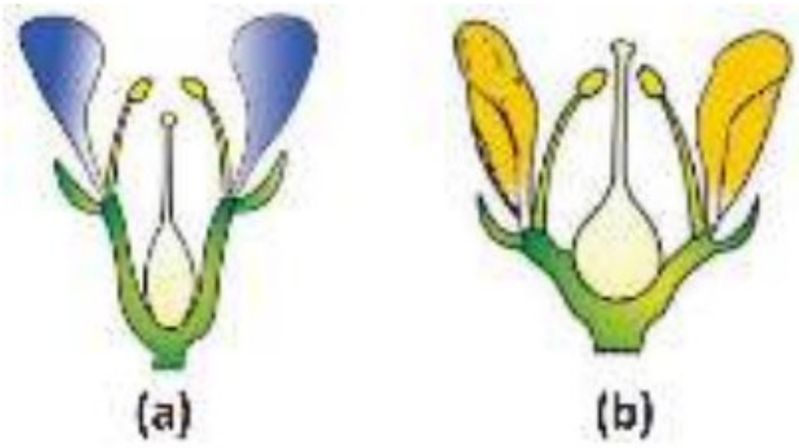
(1) Mustard
(2) Datura
(3) Chilli
(4) Gulmohur

15) Roots associated with nitrogen fixing bacteria are -

(1) Fusiform roots
(2) Napiform roots
(3) Nodulated roots
(4) Conical roots

16) Identify the type of flowers based on the position of calyx, corolla and androecium with respect to the ovary from the given figures (a) and (b).

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(1) (a)–Epigynous; (b)–Hypogynous

(2) (a)–Hypogynous; (b)–Epigynous

(3) (a)–Perigynous; (b)–Epigynous

(4) (a)–Perigynous; (b)–Perigynous

17) Papilionaceous (vexillary) aestivation consists of five petals. The largest petal is called:

(1) Keel

(2) Wing

(3) Standard

(4) Corona

18) Stamens attached to the perianth segments (as seen in Lily) are described as:

(1) Epipetalous

(2) Epiphyllous

(3) Monoadelphous

(4) Polyadelphous

19) *Citrus* plants exhibit stamens united into more than two bundles. This condition is called:

(1) Monoadelphous

(2) Diadelphous

(3) Polyadelphous

(4) Syngenesious

20) Apocarpous gynoecium (free carpels) is present in:

(1) Mustard

(2) Tomato

(3) Lotus

(4) Brinjal

21) The Floral Diagram represents which one of the following families ?

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(1) Fabaceae

(2) Brassicaceae

(3) Solanaceae

(4) Liliaceae

22) A fruit developed without fertilisation of the ovary is termed:

(1) Parthenocarpic fruit

(2) Aggregate fruit

(3) Composite fruit

(4) False fruit

23) Mango and coconut fruits are botanically classified as:

(1) Berry

(2) Capsule

(3) Drupe

(4) Nut

24) Match **List - I** with **List - II** :-

List - I	**List - II**
(a) Imbricate	(i) *Calotropis*
(b) Valvate	(ii) *Cassia*
(c) Vexillary	(iii) Cotton
(d) Twisted	(iv) Bean

Choose the **correct answer** from the options given below :

(1) (a) - (ii), (b) - (i), (c) - (iii), (d) - (iv)

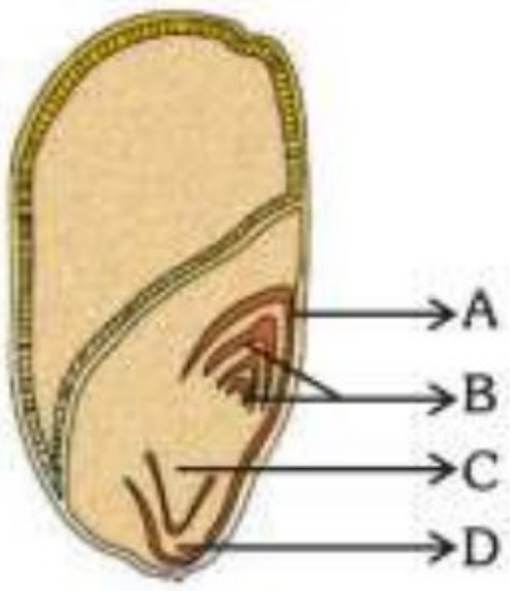
(2) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

(3) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

(4) (a) - (i), (b) - (iii), (c) - (iv), (d) - (ii)

25) Identify the part of the seed from the given figure which is destined to form root when the seed germinates.

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(1) A
(2) B
(3) C
(4) D

26) In a dicot root, the vascular cambium originates from:

(1) Epiblema and cortex
(2) Portions of pericycle tissue located above protoxylem and tissue below phloem bundles
(3) Endodermis and hypodermis
(4) Medullary rays and pith

27) In a dicot stem, the interfascicular cambium develops from the cells of:

(1) Medullary rays
(2) Pericycle
(3) Endodermis
(4) Intrafascicular cambium

28) The chief function of a xylem vessel in a plant body is to -

(1) Conduct sap
(2) Conduct mineral salts only
(3) Eliminate excess of water at night
(4) Translocate organic nutrients

29) Passage cells are thin-walled endodermal cells present opposite to:

(1) Phloem elements
(2) Protoxylem vessels
(3) Pericycle cells
(4) Secondary cortex

30) Identify the correct sequence of layers in a dicot stem from the outside to the inside:

(1) Epidermis → Endodermis → Hypodermis → Pericycle → Vascular bundle

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(2) Epidermis → Hypodermis →General Cortex →Endodermis → Pericycle
(3) Epidermis → Pericycle → Endodermis → Cortex →Medullary ray
(4) Epiblema → Cortex → Pericycle → Endodermis → Vascular bundle

31) What makes the endodermal cells of a root impermeable to water?

(1) Deposition of pectin in the outer cell wall
(2) Lignification of all cell surfaces
(3) Deposition of suberin in the form of Casparian strips on radial and tangential walls
(4) Presence of a thick cuticular layer

32) In a conjoint vascular bundle, the phloem is typically located:

(1) Inner to xylem
(2) Outer to xylem
(3) On both sides of xylem in equal measure
(4) Alternating radially with xylem

33) Closed vascular bundles lack the ability to form secondary tissues because:

(1) They lack xylem vessels
(2) Cambium is absent between xylem and phloem
(3) They are surrounded by an endodermal sheath
(4) Phloem parenchyma is present

34) Sugar transport elements of gymnosperms & pteridophytes are -

(1) Sieve cells
(2) Sieve elements
(3) Sieve tubes
(4) Sieve tube elements

35) How do bulliform cells help leaves during drought conditions?

(1) They secrete a thick waxy cuticle
(2) They become flaccid due to water loss, causing the leaves to curl inward
(3) They close the stomatal apertures directly
(4) They convert into sclerenchyma cells to reduce transpiration

36) Anatomically, a monocot stem differs from a dicot stem in having:

(1) A ring of open vascular bundles
(2) Sclerenchymatous bundle sheath around scattered vascular bundles
(3) Collenchymatous hypodermis
(4) Well-developed pith and medullary rays

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37) Which of the following is observed in sunflower plant ?

(1) 

(2) 

(3) 

(4) 

38) In a dicot root, the exarch condition of xylem means:

(1) Protoxylem lies toward the center and metaxylem toward the periphery
(2) Protoxylem lies toward the periphery and metaxylem toward the center
(3) Xylem surrounds phloem completely
(4) Cambium lies between protoxylem and metaxylem

39) Water cavities in the vascular bundles of a monocot stem are formed by the breakdown of:

(1) Metaxylem
(2) Phloem parenchyma
(3) Protoxylem elements
(4) Sclerenchyma sheath

40) Isobilateral leaves are characterized by:

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(1) Equal distribution of stomata on both surfaces and undifferentiated mesophyll

(2) Stomata present only on the abaxial surface

(3) Differentiated palisade and spongy parenchyma

(4) Reticulate venation with variable vascular bundle sizes

41) Consider the following two statements regarding leaf anatomy:
Statement I: In a dorsiventral leaf, the vascular bundles are surrounded by a layer of thick-walled bundle sheath cells.
Statement II: Vascular bundle sizes in a dicot leaf are uniform throughout the lamina regardless of vein diameter.
Choose the correct option:

(1) Both Statement I and Statement II are correct

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are incorrect

42) Consider the following two statements:
Statement I: Pericycle cells of dicot roots give rise to both lateral roots and a portion of vascular cambium.
Statement II: Monocot roots undergo limited secondary growth during severe dry seasons.
Choose the correct option:

(1) Statement I is correct, Statement II is incorrect

(2) Statement I is incorrect, Statement II is correct

(3) Both statements are correct

(4) Both statements are incorrect

43) Consider the following two statements:
Statement I: Radial vascular bundles are common in stems, while conjoint vascular bundles are common in roots.
Statement II: In radial vascular bundles, xylem and phloem are arranged on alternate radii.
Choose the correct option:

(1) Both Statements are correct

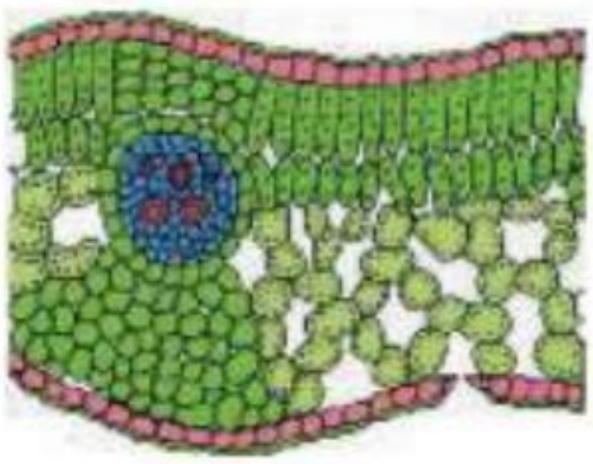
(2) Both Statements are incorrect

(3) Statement I is incorrect, Statement II is correct

(4) Statement I is correct, Statement II is incorrect

44) Identify the given diagrammatic sketch and select the correct option in respect of diagram?

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(1) Hypostomatic leaf

(2) Amphistomatic leaf

(3) Found in monocot plant

(4) Presence of bulliform cells towards abaxial surface.

45) **Assertion (A):** Peripheral vascular bundles in a monocot stem are smaller and more densely arranged than central ones.
Reason (R): Central vascular bundles are surrounded by collenchymatous sheaths while peripheral ones have parenchymatous sheaths.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

46) **Assertion (A):** The endodermis of a root regulates the entry and types of solutes that reach the xylem.
Reason (R): Root endodermal cells contain suberized Casparian strips that force water and solutes to move symplastically through cell membranes.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

47) **Assertion (A):** Phloem parenchyma is absent in most monocotyledons.
Reason (R): Monocot leaves lack veins and companion cells.

(1) Both A and R are true

(2) Both A and R are false

(3) A is true but R is false

(4) A is false but R is true

48) Which of the following anatomical features differentiate a monocot root from a dicot root?
I. Presence of polyarch xylem condition (more than 6 patches)
II. Large and well-developed pith
III. Absence of secondary growth
IV. Presence of Casparian strips in endodermis

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(1) I, II, and III only

(2) I and II only

(3) II, III, and IV only

(4) I, III, and IV only

49) Identify the features characteristic of a dicot stem:
I. Collenchymatous hypodermis
II. Ring arrangement of vascular bundles
III. Conjoint, open vascular bundles with endarch protoxylem
IV. Presence of starch sheath (endodermis)

(1) I and II only

(2) II, III, and IV only

(3) I, II, III, and IV

(4) I, III, and IV only

50) Which of the following components are part of the ground tissue system?
I. Cortex
II. Pericycle
III. Mesophyll
IV. Epidermis

(1) I, II, and III only

(2) I and III only

(3) II, III, and IV only

(4) I, II, III, and IV

51) Potato and brinjal are placed in the same genus because they:

(1) Are identical in structure

(2) Belong to the same family only

(3) Have many common characteristics

(4) Are cultivated in the same region

52) On what basis are plant families mainly characterised?

(1) Habitat and distribution

(2) Vegetative and reproductive features

(3) Mode of nutrition

(4) Size of the plant

53) As we move from species to kingdom, what happens to common characteristics?

(1) They increase

(2) They remain the same

(3) They decrease

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(4) They disappear completely

54) Animals like fishes, amphibians, reptiles, birds and mammals are grouped under phylum Chordata because they:

(1) Live in similar habitats

(2) Have similar food habits

(3) Possess notochord and dorsal hollow neural system

(4) Belong to the same class

55) Which of the following correctly represents the taxonomic hierarchy from lower to higher category?

(1) Family → Genus → Species → Order

(2) Species → Genus → Family → Order

(3) Order → Family → Genus → Species

(4) Species → Family → Genus → Order

56) The animal order *Carnivora* includes which of the following families?

(1) Solanaceae and Convolvulaceae

(2) *Panthera* and *Felis*

(3) Mammalia and Primata

(4) Felidae and Canidae

57) **Assertion (A):** Binomial nomenclature uses two words for naming an organism.
Reason (R): The first word represents the genus and the second represents the specific epithet.

(1) Both A and R are true, and R explains A

(2) Both A and R are true, but R does not explain A

(3) A is true, R is false

(4) A is false, R is true

58) **Assertion (A):** Taxonomy includes identification, classification, nomenclature and characterisation.
Reason (R): Taxonomy deals only with naming of organisms.

(1) Both A and R are true, and R explains A

(2) Both A and R are true, but R does not explain A

(3) A is true, R is false

(4) A is false, R is true

59) Match Column I with Column II :

Column I	Column II
A. ICBN	1. Naming of animals
B. ICZN	2. Naming of plants

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C. Binomial nomenclature	3. Two-word naming system
D. *Mangifera*	4. Genus of mango

(1) A–1, B–2, C–3, D–4

(2) A–2, B–3, C–1, D–4

(3) A–2, B–1, C–3, D–4

(4) A–3, B–1, C–2, D–4

60) Which of the following statements about taxonomy are correct ?
1. Taxonomy involves classification and nomenclature.
2. Systematics considers evolutionary relationships.
3. Classification is based only on habitat.
4. Taxa represent different levels of classification.
5. Species is the highest taxonomic category.

(1) 1, 2 and 4

(2) 2, 3 and 5

(3) 1, 3 and 5

(4) 1, 2, 3 and 4

61) Consider the following statements regarding species -
1. Species is the lowest taxonomic category.
2. Members of a species show fundamental similarities.
3. Species can be identified only by reproductive isolation.
4. Morphological differences help distinguish closely related species.
Which of the above statements are correct?

(1) 1 and 3 only

(2) 1, 2 and 4 only

(3) 2 and 3 only

(4) 1, 2, 3 and 4

62) The scientific name *Mangifera indica Linn.* conveys which of the following information?

(1) Mango belongs to genus Indica

(2) Linn was the first person to discover mango

(3) Linn was the first to describe the species scientifically

(4) Mango was first found in India

63) Which of the following pairs is incorrectly matched?

(1) ICBN – Botanical nomenclature

(2) ICZN – Zoological nomenclature

(3) Systematics – Study of evolutionary relationships

(4) Taxonomy – Study of economic uses of organisms

64) **Assertion (A):** Systematics deals with the study of diversity of organisms and their relationships.

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Reason (R): Systematics takes into account evolutionary relationships between organisms.

(1) Both A and R are true, and R explains A

(2) Both A and R are true, but R does not explain A

(3) A is true, R is false

(4) A is false, R is true

65) **Assertion (A):** The term systematics is derived from the Latin word “systema”.
Reason (R): The word “systema” means systematic arrangement of organisms.

(1) Both A and R are true, and R explains A

(2) Both A and R are true, but R does not explain A

(3) A is true, R is false

(4) A is false, R is true

66) Which of the following **correctly matches** embryonic development and classification?

Phylum	**Germ Layers**	**Coelom Status**	**Notochord**
(1) Mollusca	Diploblastic	Coelomate	Absent
(2) Echinodermata	Triploblastic	Coelomate	Present
(3) Cnidaria	Diploblastic	Acoelomate	Absent
(4) Platyhelminthes	Triploblastic	Coelomate	Absent

Choose the correct set:

(1) a, b, and c

(2) a, c, and d

(3) a and c only

(4) a and b only

67) Which of the following are correct regarding *Periplaneta americana*?
1. Belongs to largest phylum of animal kingdom.
2. Excretion through nephridia.
3. Open circulatory system.
4. Radially symmetrical.
5. Uricotelic mode of excretion.

(1) 1, 3, and 5

(2) 2, 3, and 4

(3) 1, 2, and 5

(4) 1, 4, and 5

68) Consider the following statements about Annelida and Arthropoda**:**
1. Both show metameric segmentation.
2. Arthropods have an open circulatory system, annelids have a closed one.
3. Excretion occurs via nephridia in annelids and Malpighian tubules in Arthropods.
4. Arthropods have jointed Appendages, annelids do not.
Choose the correct set of statements:

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(1) 1, 2, and 3 only

(2) 1, 2, 3, and 4

(3) 2, 3, and 4 only

(4) 1 and 4 only

69) Given a sponge showing ostia, spongocoel, and osculum, identify the correct sequence of water flow.

(1) Osculum → Spongocoel → Ostia

(2) Ostia → Spongocoel → Osculum

(3) Spongocoel → Ostia → Osculum

(4) Ostia → Osculum → Spongocoel

70) Arrange the following phyla according to their type of **nervous system** from simplest to most complex:
Porifera, Platyhelminthes, Annelida, Arthropoda

(1) Porifera → Platyhelminthes → Annelida → Arthropoda

(2) Porifera → Annelida → Platyhelminthes → Arthropoda

(3) Platyhelminthes → Porifera → Annelida → Arthropoda

(4) Porifera → Platyhelminthes → Arthropoda → Annelida

71) How many of the following are correct descriptions?
1. *Physalia* (Portuguese man-of-war) is a cnidarian that exhibit tissue level of organisation and are diploblastic.
2. *Ascaris* is an acoelomate roundworm with a complete digestive tract and longitudinal muscle movement.
3. Earthworm has a closed circulatory system, and is monoecious.
4. *Starfish* shows tube feet-based locomotion, water vascular system, and reproduction is sexual.
5. *Tapeworms* reproduce sexually through separate sexes and develop via trochophore larva.

(1) Only 4 statements are correct

(2) Only 3 statements are correct

(3) All 5 statements are correct

(4) Only 2 statements are correct

72) With calcareous plates below my skin,
I show symmetry that spins and spins.
Water vascular flow is how I move,
And tube feet are my locomotion groove."
Who am I?

(1) *Sycon*

(2) *Asterias*

(3) *Pinctada*

(4) *Balanoglossus*

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73) Metagenesis refers to :

(1) Presence of a segmented body and parthenogenetic mode of reproduction.

(2) Presence of different morphic forms.

(3) Alternation of generation between asexual and sexual phases of an organism.

(4) Occurrence of a drastic change in form during post-embryonic development.

74) Which of the following statements is/are correct?
1. Notochord is always replaced by vertebral column in vertebrates.
2. All chordates exhibit pharyngeal slits in embryonic development.
3. Cephalochordates lack a true brain.
4. Urochordates are more advanced than cephalochordates in terms of chordate features.

(1) 1, 2, and 3 only

(2) 2 and 3 only

(3) 1, 2, 3, and 4

(4) 2 and 4 only

75)

(A) Sponges (Phylum Porifera) are asymmetrical animals that possess porous bodies with specialized cells called choanocytes which help in generating water current and capturing food particles through intracellular digestion.
(B) Cnidarians exhibit radial symmetry and diploblastic organization with two germ layers, ectoderm and endoderm, and possess specialized stinging cells called cnidocytes used for capturing prey and defense.
(C) Poriferans have a well-developed nervous system and exhibit coordinated movement through their contractile tissues that surround the osculum for regulating water flow.
(D) In cnidarians, the body has a gastrovascular cavity with a single opening that serves as both mouth and anus, and they exhibit two major body forms — polyp and medusa, which show alternation during their life cycle.

	A	**B**	**C**	**D**
(1)	True	True	False	True
(2)	True	False	True	True
(3)	False	True	False	False
(4)	True	True	True	True

(1) 1

(2) 2

(3) 3

(4) 4

76) Fill in the blanks:

Animal	**Phylum**	**Coelom Type**	**Excretory Organ**	**Segmentation**

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Earthworm	___**(1)**___	___**(2)**___	___**(3)**___	Metameric
Ascaris	Nematoda	___**(4)**___	Renette cells	___**(5)**___

(1) (1) Annelida, (2) True coelom, (3) Nephridia, (4) Pseudocoelomate, (5) Unsegmented

(2) (1) Arthropoda, (2) Pseudocoelomate, (3) Flame cells, (4) True coelom, (5) Segmented

(3) (1) Annelida, (2) Acoelomate, (3) Green glands, (4) Pseudocoelomate, (5) Metameric

(4) (1) Mollusca, (2) True coelom, (3) Flame cells, (4) Acoelomate, (5) Segmented

77) A zoologist observes an animal with the following traits:
 Bilaterally symmetrical, unsegmented body.
 Pseudocoelomic body cavity.
 Complete alimentary canal with well developed muscular pharynx.
 Males and females are distinct.
What phylum does this animal belong to? How does its excretory system function?

(1) Nematoda; excretion through protonephridia

(2) Annelida; excretion through nephridia

(3) Nematoda; excretion via renette cells

(4) Platyhelminthes; excretion through flame cells

78) As radula is to *Pila*, so is proboscis to _____ ?

(1) *Balanoglossus*

(2) *Ascaris*

(3) Leech

(4) *Sycon*

79) In which animal is the notochord confined only to the tail region of the larva, and completely absent in adults?

(1) *Branchiostoma*

(2) *Petromyzon*

(3) *Herdmania*

(4) *Myxine*

80) As radula is to snail, choanocytes are to _____ ?

(1) Hydra

(2) Sponge

(3) Earthworm

(4) Starfish

81) Which of the following is a correct combination of characters related to the given animal?



(1) Jaws present, paired fins absent, bony skeleton

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(2) Jaws absent, paired fins absent, cartilaginous skeleton
(3) Jaws absent, paired fins present, cartilaginous skeleton
(4) Jaws present, paired fins present, cartilaginous skeleton

82) In Osteichthyes, the gills are:

(1) Three pairs and uncovered
(2) Four pairs and covered by operculum
(3) Five pairs and without operculum
(4) Four pairs and exposed directly to water

83) Which one differs from the other three with respect to **level of organisation**?

(1) *Sycon*
(2) Hydra
(3) *Pleurobrachia*
(4) *Ctenoplana*

84) I am an animal with:
Tissue-level organisation
Radial symmetry
Diploblastic body
Gastrovascular cavity
Specialised stinging cells
Who am I?

(1) *Pleurobrachia*
(2) Hydra
(3) Sycon
(4) *Planaria*

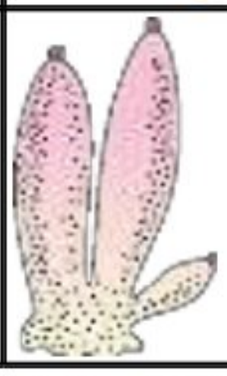
85) **Organism X:** Notochord extends from head to tail region and persists throughout life.
Organism Y: Notochord is replaced by a vertebral column during development.
Which option correctly identifies them?

(1) X - Urochordata, Y - Cephalochordata
(2) X - Cephalochordata, Y - Vertebrata
(3) X - Vertebrata, Y - Urochordata
(4) X - Hemichordata, Y - Cephalochordata

86) Choose the correct match from column-I, II and III :-

Column-I	**Column-II**	**Column-III**

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A		i	Coelenterata	a	Tissue level of organisation
B		ii	Aschelminthes	b	Organ system level of organisation
C		iii	Porifera	c	Organ/organ system level of organisation
D		iv	Platyhelminthes	d	Cellular level of organisation

(1) A-i-a, B-ii-b, C-iii-c, D-iv-d

(2) A-ii-b, B-i-a, C-iv-c, D-iii-d

(3) A-iii-c, B-i-d, C-ii-a, D-iv-a

(4) A-iv-d, B-iii-c, C-iv-b, D-i-b

87) Identify the incorrect statement about the given organism :-



(1) Body in divided into proboscis, collar & trunk.

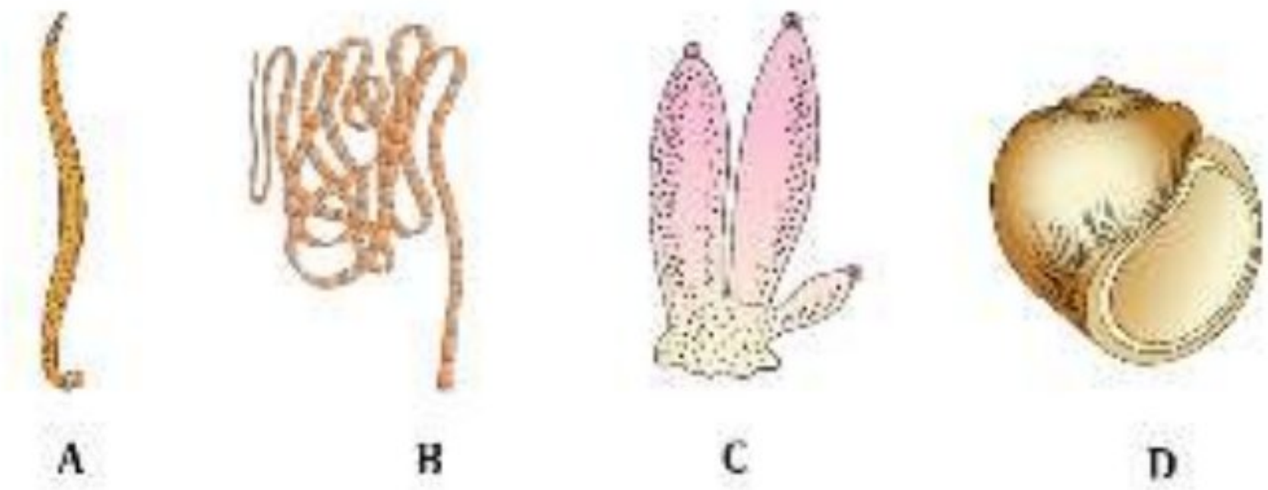
(2) Bilaterally symmetrical, diploblastic and coelomate animals.

(3) Respiration takes place through gills.

(4) Excretory organ is proboscis gland.

88) Identify the animals given in figure A, B, C and D -

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(1) A - *Ascidia*, B - *Sycon*, C - *Ascaris*, D - *Asterias*

(2) A - *Ascaris*, B - Tapeworm, C - *Sycon*, D - *Pila*

(3) A - Tapeworm, B - *Ascaris*, C - *Pila*, D - *Sycon*

(4) A - *Sycon*, B - *Pila*, C - *Ascaris*, D - Tapeworm

89) 

Identify the statements which are true to the given animal:

(a) Body divisible into head and trunk.

(b) Moist skin without scales.

(c) Eyes do not have eyelids.

(d) Cloaca is absent.

(1) (a) and (d)

(2) (b) and (c)

(3) (c) and (d)

(4) (a) and (b)

90) Which of the following bear 6–15 pairs of gill slits for respiration ?

(1) 

(2) 

(3) 

(4) 

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**3**	**2**	**2**	**4**	**4**	**4**	**1**	**2**	**2**	**1**	**2**	**1**	**3**	**1**	**1**	**1**	**2**	**2**	**1**	**4**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**1**	**3**	**1**	**1**	**1**	**1**	**4**	**1**	**1**	**4**	**2**	**4**	**4**	**1**	**3**	**3**	**3**	**2**	**3**	**1**
Q.	41	42	43	44	45															
A.	**2**	**4**	**4**	**2**	**1**															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	**1**	**1**	**4**	**4**	**3**	**2**	**2**	**4**	**3**	**2**	**4**	**4**	**4**	**2**	**4**	**2**	**4**	**1**	**3**	**1**
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	**2**	**2**	**2**	**4**	**1**	**1**	**2**	**3**	**2**	**2**	**2**	**2**	**4**	**2**	**2**	**4**	**3**	**3**	**4**	**1**
Q.	86	87	88	89	90															
A.	**3**	**1**	**2**	**1**	**4**															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	**3**	**4**	**3**	**1**	**1**	**1**	**1**	**2**	**4**	**1**	**3**	**3**	**3**	**4**	**3**	**4**	**3**	**2**	**3**	**3**
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	**2**	**1**	**3**	**2**	**3**	**2**	**1**	**1**	**2**	**2**	**3**	**2**	**2**	**1**	**2**	**2**	**1**	**2**	**3**	**1**
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**2**	**1**	**3**	**1**	**3**	**1**	**3**	**1**	**3**	**1**	**3**	**2**	**3**	**3**	**2**	**4**	**1**	**3**	**3**	**1**
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**2**	**3**	**4**	**1**	**1**	**3**	**1**	**2**	**2**	**1**	**2**	**2**	**3**	**2**	**1**	**1**	**3**	**1**	**3**	**2**
Q.	171	172	173	174	175	176	177	178	179	180										
A.	**2**	**2**	**1**	**2**	**2**	**2**	**2**	**2**	**4**	**1**										

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SOLUTIONS

PHYSICS

1) **Explaining :** A wire *PQRS* carrying a current *I* runs along three edges of a cube of side as shown. There exists a uniform magnetic field of magnitude B along one of the sides of the cube. The magnitude of the force acting on the wire is

Concept : This question is based on Force on current-carrying conductor in B-field

Solution : Force on the part of wire which is parallel to $\vec{B}$ will be zero. The forces on the other two parts will each be IlB in perpendicular directions.

Final Answer : (3)

2) **Explaining :** Consider the inferences given below in respect of the following current loop of wire kept in a magnetic field $\vec{B}$

(1) The magnitude and direction of force on the element AC of the wire is $\frac{\sqrt{3}}{2}ILB$ directed into the page

(2) The magnitude and direction of force on the element AB of the wire is $\frac{\sqrt{3}}{2}ILB$ directed into the page

(3) The total force on the loop ABCA is zero

Which of the above is/are not true

Concept : This question is based on $F = I\ell B\sin\theta$.

Solution : Force on the element AC of the wire is

$$F_{AC} = I(L\cos 30^\circ)B\sin 90^\circ = \frac{\sqrt{3}}{2}ILB$$

It is directed into the page.

 Statement 1 is true.

Force on the element AB of the wire is

$$F_{AB} = ILB\sin(90^\circ - 30^\circ) = ILB\sin 60^\circ = \frac{\sqrt{3}}{2}ILB$$

It is directed out of the page.

 Statement 2 is not true.

Total force on the current carrying loop in a uniform magnetic field is zero.

 Total force on the loop ABCA is zero.

Statement 3 is true.

Final Answer : (2)

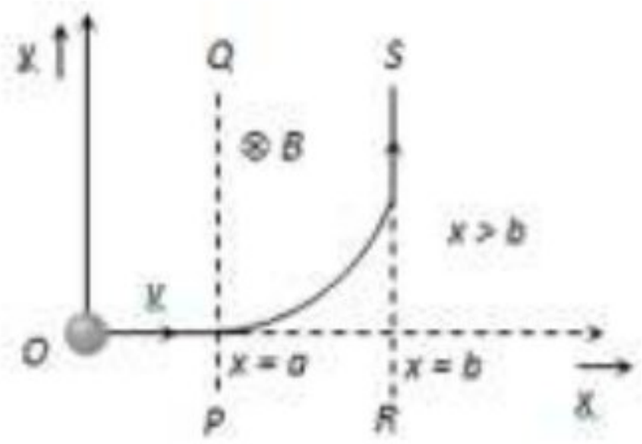
3) **Explaining :** A particle of mass m and charge q moves with a constant velocity v along the positive x direction. It enters a region containing a uniform magnetic field B directed along the negative z direction, extending from x = a to x = b. What is the value of v required so that the particle can just enter the region x > b is

Concept : This question is based on r > b – a

condition for entering the region x > b

Solution : In the figure, the z-axis points out of the paper, and the magnetic field is directed into the paper, existing in the region between PQ and RS. The particle moves in a circular path of radius r in the magnetic field. It can just enter the region x > b for r > (b – a)

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$r > (b - a)$

$$\frac{mv}{qB} > b - a$$

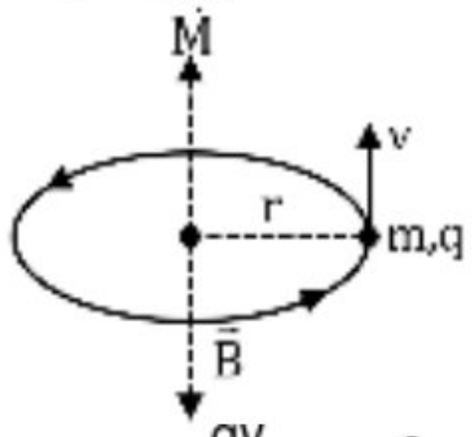
$$v > \frac{q(b-a)B}{m}$$

Final Answer : (2)

4) **Explaining :** To Choose the correct graph for diamagnetic materials.
Concept : This question is based on magnetic materials.
Solution : For diamagnetic materials, (□) is independent of temperature and M ∝ H but M is in opposite direction of H.
Final Answer : (4)

5) **Explain Question :** To find the magnetic moment of moving particle.
Concept : atomic magnetism
Solution :

$$I = \frac{q}{t} = \frac{qv}{2\pi r}$$



$$M = IA = \frac{qv}{2\pi r} \times \pi r^2$$

$$= \frac{1}{2}qvr = \frac{1}{2}qv \times \frac{mv}{qB}$$

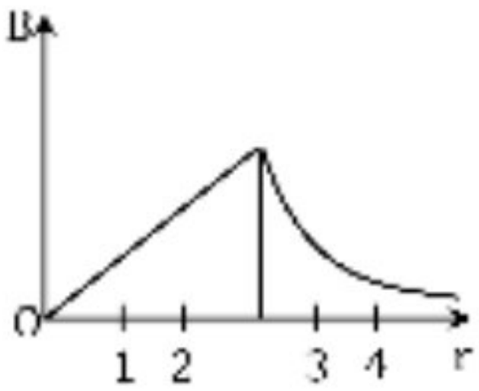
$$= \frac{mv^2}{2B}$$

Final Answer : Option **(4)**

6) **Explaining :** The adjacent figure shows the cross-section of a long rod with its length perpendicular to the plane of the paper. It carries a constant current flowing along its length. B_1, B_2, B_3 and B_4 respectively represent the magnetic fields due to the current in the rod at point 1, 2, 3 and 4 lying at different separations from the centre O, as shown in the figure. Which of the following shall hold true
Concept : This question is based on $B_{in} \propto r$.

$$B_{out} \propto \frac{1}{r}$$

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Solution :

Final Answer : (4)

7)

Question Explanation: In this question, a square current-carrying coil is placed near a long straight current-carrying wire. We are asked to find the net force on the coil and its direction of motion.

Concept : This question is based on Force on a wire/loop in non uniform magnetic field.

Solution : The loop is located in non uniform magnetic field

So force on elemental length of the horizontal wire will be of same magnitude and opposite direction therefore net force on horizontal wire will be zero

• Net force is due to vertical sides only

• Left vertical side (nearer) distance

$r_1 = 2cm = 0.02$ m

• Right vertical side (farther)

$r_2 = 2 + 10 = 12$ cm $= 0.12$ m

• Length of vertical side:

l = 15 cm = 0.15 cm

$$F_{near} = \frac{\mu_0 I_1 I_2 l}{2\pi r_1}$$

$$F_{far} = \frac{\mu_0 I_1 I_2 l}{2\pi r_2}$$

 $r_1 < r_2$, \ $F_{near} > F_{far}$

 Net force

$F_{net} = F_{near} - F_{far}$

$$F_{net} = \frac{\mu_0 I_1 I_2 l}{2\pi}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

Substituting values:

$I_1 = 2_A$, $I_2 = 1A$

$$F_{net} = \frac{4\pi \times 10^{-7}}{2\pi} \times 2 \times 1 \times 0.15\left(\frac{1}{0.02} - \frac{1}{0.12}\right)$$

$$F_{net} = 2 \times 10^{-7} \times 0.3 \times (50 - 8.33) \approx 2.5 \times 10^{-6} N$$

$= 25 \times 10^{-7}$ n

Direction of force

• Current in straight wire and near vertical side are in same direction

• Same direction currents = attraction

• Near side force dominates ⇒ coil moves towards the wire

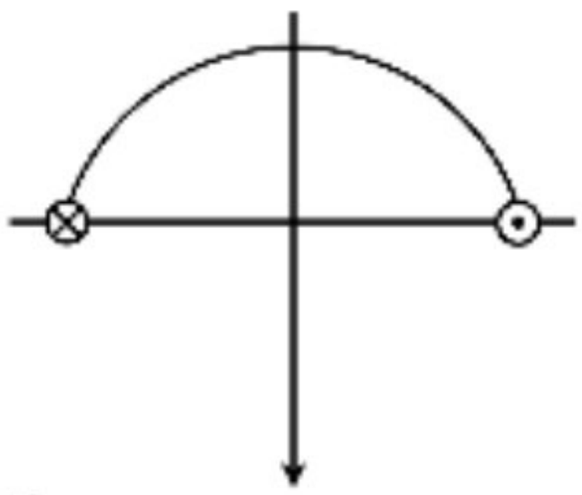
Final Answer : Option (1)

8) **Explanation :** Find MF due to this configuration

Concept : This question is based on use formula

Solution :

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$\vec{B}_1 \rightarrow$ Net magnetic field due to semi infinite wire

$\vec{B}_2 \rightarrow$ magnetic field due to semi circular wire

$$\vec{B}_1 = \frac{2\mu_0 I}{4\pi R}$$

$$\vec{B}_2 = \frac{\mu_0 I}{4R}$$

So, $\vec{B} = \frac{\mu_0 I}{4\pi R}(-2\hat{k} - \pi\hat{i})$

Final Answer : (2)

9) **Explaining :** The value of $\oint \vec{B} \cdot d\vec{\ell}$ around the closed loop will be

Concept : This question is based on $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enclosed}$

Solution : Loop orientation is clockwise

$\therefore \otimes \rightarrow$ Positive $\odot \rightarrow$ Negative

Final Answer : (2)

10) **Explaining :** A long solenoid has 200 turns per cm and carries a current i. The magnetic field at its centre is 6.28×10^{-2} weber/m^2. Another long solenoid has 100 turns per cm and it carries a current i/3. The value of the magnetic field at its centre is-

Concept : This question is based on $B = \mu_0 ni$.

Solution : $B_{solenoid} = \mu_0 ni$

$$\frac{B_2}{B_1} = \frac{n_2 i_2}{n_1 i_1} = \frac{100 \times i/3}{200 \times i}$$

$$B_2 = \frac{B_1}{6} = \frac{6.28 \times 10^{-2}}{6}$$

$= 1.05 \times 10^{-2}$ T

Final Answer : (1)

11) **Explanation :** Find MF at the centre of the solenoid

Concept : This question is based on $B = \mu_0 nI$

Solution :

$$B = \mu_0 \frac{N}{\ell} I$$

$$= 4\pi \times 10^{-7} \times \frac{100}{(0.5)} \times 2.5$$

$= 6.28 \times 10^{-4}$ T

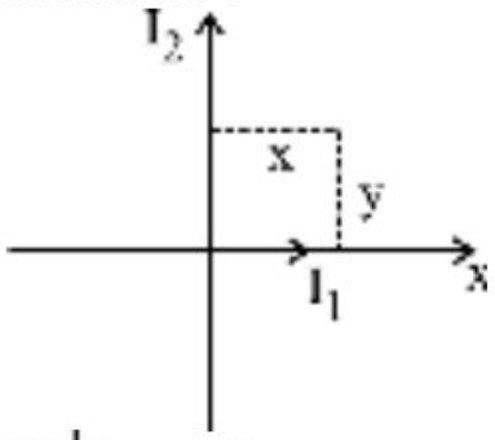
Final Answer : (2)

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12) **Explanation :** Find relation b/w x & y coordinate.

Concept : This question is based on $B_1 = B_2$

Solution :



$$\frac{\mu_0 I_1}{2\pi y} = \frac{\mu_0 I_2}{2\pi x}$$

$$\Rightarrow y = \frac{I_1}{I_2} x$$

Final Answer : (1)

13) **Explanation :** find $\frac{B}{B'}$

Concept : This question is based on use formula for cylinder

Solution :

For points inside the wire, $B = \frac{\mu_0 Ir}{2\pi R^2} (r \leqslant R)$

For points outside the wire, $B = \frac{\mu_0 I}{2\pi r} (r \geqslant R)$

$$\frac{B}{B'} = \frac{\frac{\mu_0 I(a/2)}{2\pi a^2}}{\frac{\mu_0 I}{2\pi(2a)}} = 1:1$$

according to the question,

Final Answer : (3)

14) **Explanation :** Find MF due to toroid

Concept : This question is based on $B = \mu_0 \frac{N}{2\pi R} I$

Solution :

$$B = \mu^0 ni = \frac{\mu_0 N}{2\pi R_{Mean}} \times i$$

$$= \frac{4\pi \times 10^{-7} \times 240 \times 2}{2 \times \pi \times 20 \times 10^{-2}} = 4.8 \times 10^{-4}\ T$$

Final Answer : (1)

15) **Explaining :** find magnetic field due to Arc

Concept :

$$B = \frac{\mu_0 I}{4\pi r}\theta$$

Solution :

'B' due to outer Arc $\Rightarrow B_1 = \left(\frac{\mu_0 i\theta}{4\pi b}\right) \otimes$

'B' due to inner arc $\Rightarrow B_1 = \left(\frac{\mu_0 i\theta}{4\pi a}\right) \odot$

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Net 'B' at centre 'O' = $|B_2 - B_1|\odot$

$= \left(\frac{\mu_0 i\theta}{4\pi}\right)\left[\frac{1}{a} - \frac{1}{b}\right]\odot$

$= \frac{\mu_0 i\theta}{4\pi}\left[\frac{b-a}{ab}\right]\odot$

Final Answer : (1)

16) **Explaining :** find radius of circular path

Concept :

$R = \frac{mv}{qB}$

Solution :

$R = \frac{mV}{qB} = \frac{V}{\frac{q}{m}B} = \frac{4 \times 10^5}{5 \times 10^7 \times 8 \times 10^{-2}} = 0.1\ m$

Final Answer : (1)

17) **Explanation :** Identify the particles

Concept : This question is based on $r = \frac{mv}{qB}$

Solution :

$r = \frac{mv}{qB} = \frac{p}{qB}$

$r \propto \frac{1}{q}$

$q_p = q_d < q_\alpha$

$r_{proton} = r_{deutron} > r_{\alpha\text{-particle}}$

Final Answer : (2)

18) **Explaining :** find force between two current carrying parallel wire

Concept :

$F = \frac{\mu_0 I_1 I_2}{2\pi d}\ell$

Solution :

$F = \frac{\mu_0 I_1 I_2 \ell}{2\pi d}$

$= \frac{4\pi \times 10^{-7} \times 10 \times 2 \times 2}{2\pi \times 10 \times 10^{-2}}$

$F = 8 \times 10^{-5}\ N$

Final Answer : (2)

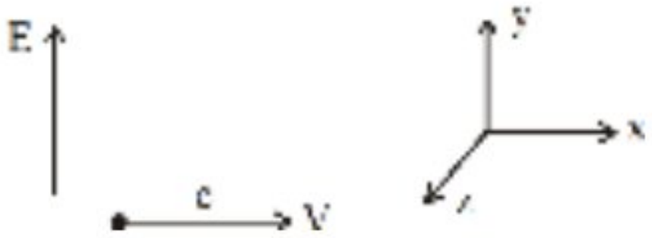
19)

Question Explanation: We need a magnetic field so the electron continues straight along +x.

Concept: Velocity selector: qE=qvB

Solution :

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$\vec{B}$ must be in +z axis.

$\vec{V} = 6 \times 10^6 \hat{i}$

$\vec{E} = 300\hat{j}$ V/cm $= 3 \times 10^4$ V/m

$q\vec{E} + q\vec{V} \times \vec{B} = 0$

$E = VB$

$$B = \frac{E}{V} = \frac{3 \times 10^4}{6 \times 10^6} = 5 \times 10^{-3} T$$

Final Answer : Option (1)

20) **Explaining :** A charged particle is projected in uniform magnetic field B with velocity v, having charge q perpendicular to the magnetic field. Then the force on charged particle is given by $F = q\left(\vec{v} \times \vec{B}\right)$.

Concept :

$$\omega = \frac{qB}{m}$$

Solution :

Remains constant whatever may be velocity of projection of given charged particle $\omega = \frac{qB}{m}$

Final Answer : (4)

21) **Explaining : Column - I** represent a charge particle being thrown in electric field and magnetic field combination. **Column - II** represent the path followed by charge particle.

Concept :

$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$

Solution :

A) $F_m = 0$, $F_E = qE$; straight line

B) $F_m = qvB$, $F_E = 0$; circular path

C) $F_m = qvB \sin\theta$, $F_E = 0$; Helix

D) $a = \frac{qE}{m} \& v$ have different angle

thus particle perform parabola

Final Answer : (1)

22) **Explanation :** Identify path

Concept : This question is based on $\vec{v} \perp \vec{B}$ then particle perform circular motion

Solution :

$\vec{v} \perp \vec{B}$ thus electron perform circular path

Final Answer : (3)

23) **Explanation :** Find lorentz force

Concept : This question is based on $\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$

Solution :

$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$

$\vec{F} = 2 \times 2\hat{j} + 2[(2\hat{i} + 3\hat{j}) \times 5\hat{k}]$

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$\vec{F} = 4\hat{j} + 2[10(-\hat{j}) + 15\hat{i}]$
$\vec{F} = 4\hat{j} - 20\hat{j} + 30\hat{i}$
$= -16\hat{j} + 30\hat{i}$
Final Answer : (1)

24) **Explanation :** Find force on charge

Concept : This question is based on $\vec{F} = q\left(\vec{V} \times \vec{B}\right)$
Solution :
$\vec{F}_B = q\left(\vec{V} \times \vec{B}\right) = +\left(\hat{j} \times -\hat{k}\right) = -\hat{i}$
Final Answer : (1)

25) **Question Explanation:**
Distance between helical turns.

Concept:
Helical Motion Pitch.

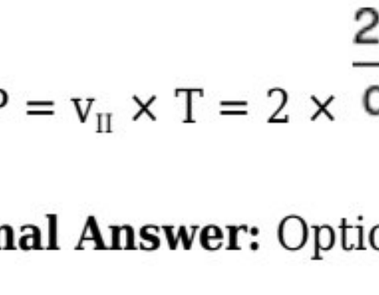
Solution:
1. Velocity components: Parallel velocity $u_{||}$ = 2 m/s (along B) and perpendicular velocity v⊥ = 3 m/s.

2. Time Period (T): The time for one circular revolution is T = $\frac{2\pi m}{qB}$
3. Pitch (P): The linear distance moved along B in one time period:

$$P = v_{II} \times T = 2 \times \frac{2\pi m}{q(5)} = \frac{4\pi m}{5q}$$

Final Answer: Option (1)

26) **Explaining :** To find which configuration has highest magnetic dipole moment.
Concept : This question is based on
direction of magnetic dipole moment is from south to north pole.
$M = \sqrt{M_1^2 + M_2^2 + 2M_1M_2\cos\theta}$
Solution :

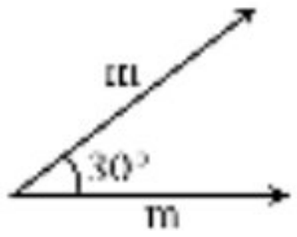
a. 

$\Rightarrow M_1 = m\sqrt{2}$

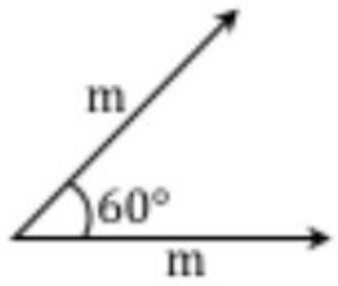
b.

$\Rightarrow M_2 = 0$

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c. 

$$M_3 = m\sqrt{2(1+\cos 30^\circ)} = m\sqrt{2\left(1+\frac{\sqrt{3}}{2}\right)} = m\sqrt{2+\sqrt{3}}$$

d. 

$M_4 = 2m\cos 30^\circ = m\sqrt{3}$

Final Answer : (1)

27) **Question Explanation:**

Find the ratio of distances from the magnet center when axial and equatorial magnetic fields are equal.

Concept:

For a short bar magnet:

A. Axial field is twice the equatorial field at same distance.

Formula:

Axial field:

$$B_{axis} = \frac{\mu_0}{4\pi}\frac{2M}{x^3}$$

Equatorial field:

$$B_{eq} = \frac{\mu_0}{4\pi}\frac{M}{y^3}$$

Calculation

Given:

$B_{axis} = B_{eq}$

$$\frac{2M}{x^3} = \frac{M}{y^3}$$

$$\frac{2}{x^3} = \frac{1}{y^3}$$

$x^3 = 2y^3$

$$\frac{x}{y} = 2^{1/3}$$

Final Answer : Option (4)

28) **Explaining :** The length of wire (or circumference of circular loop If I is current flowing in the loop and magnetic moment M.

Concept : This question is based on

M = IA

circumference of circular loop = length of wire.

Solution :

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Let L be the length of the wire. Then $L = 2\pi r$

where r is the radius of the circular loop or $r = \frac{L}{2\pi}$

Area of the loop, $A = \pi r^2 = \pi\left(\frac{L}{2\pi}\right)^2 = \frac{L^2}{4\pi}$

Magnetic moment, $M = IA = \frac{IL^2}{4\pi}$ or $L = \sqrt{\frac{4\pi M}{I}}$

Final Answer : (1)

29) **Explaining :** Given : Work done to move a magnet from $\theta = 0°$ to $\theta = 60°$ is 0.8 J in uniform magnetic field.

Ask : Work done to move magnet by 30° further.

Concept : This question is based on

$U = -\vec{M} \cdot \vec{B} = -MB\cos\theta$

$W = U_f - U_i$

Solution : $W = MB(\cos\theta_1 - \cos\theta_2)$

$0.8 = MB(\cos0° - \cos60°)$

$W = MB(\cos60° - \cos90°)$

$\frac{W}{0.8} = 1 \Rightarrow W = 0.8J$

Final Answer : (1)

30) **QUESTION EXPLANATION :**

A closed current loop PQRS carrying current is placed in a uniform magnetic field. The magnetic forces acting on segments PS, SR and RQ are F_1, F_2 and F_3 respectively, acting in the plane of the paper as shown. The force acting on segment QP is to be determined.

CONCEPT : Net magnetic force on a closed current loop in a uniform magnetic field

FORMULA :

$\vec{F}_{net} = 0$

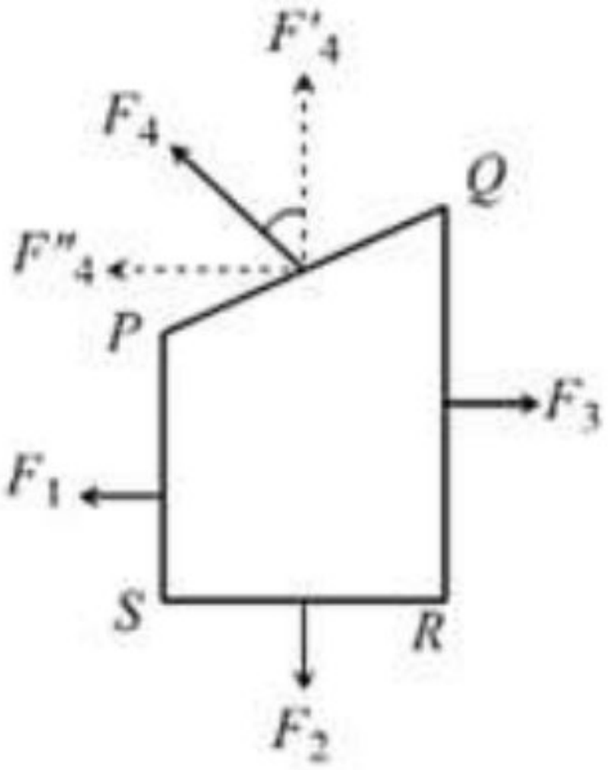
$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 = \vec{0}$

where $\vec{F}_4$ is the force on segment QP.

SOLUTION :

Since all the given forces are lying in plane, so the given loop is in equilibrium.

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From equilibrium of forces on the closed loop,

$$\vec{F}_4 = -\vec{F}_3 - \vec{F}_2 - \vec{F}_1$$

From the given directions,

$$\vec{F}_1 = -F_1\hat{i}$$
$$\vec{F}_3 = +F_3\hat{i}$$
$$\vec{F}_2 = -F_2\hat{j}$$

Substituting,

$$\vec{F}_4 = -[F_3 - F_1)\hat{i} - F_2\hat{j}]$$
$$\vec{F}_4 = -(F_3 - F_1)\hat{i} + F_2\hat{j}$$

Magnitude of force on segment QP,

$$|\vec{F}_4| = \sqrt{[(F_3 - F_1)^2 + F_2{}^2]}$$

FINAL ANSWER : Option (4)

31) **Explain the Question :** A flywheel speeds up from 20π to 40π rad/s in 10s; find number of rotations.

Concept : This question is based on Kinematics of circular motion.

Solution :

$$\theta = \frac{(\omega_i + \omega_f)\,t}{2}$$

$$2\pi N = \frac{(20\pi + 40\pi)}{2}10$$

N = 150

Final Answer : (2)

32)

Explain : how the velocity and acceleration of a particle behave when it moves in a circle with a uniform speed.

Concept : uniform circular motion.

Solution : Velocity: Velocity is a vector quantity with both magnitude (speed) and direction.

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While the magnitude (speed) is constant, the direction of the velocity vector (tangent to the circle) changes continuously as the particle moves along the circular path.

Acceleration: Acceleration is also a vector quantity. In uniform circular motion, the acceleration (centripetal acceleration) is directed towards the center of the circle. Its magnitude is constant ($a = v^2/r$), but its direction changes continuously as the particle moves along the circular path.
Therefore, both velocity and acceleration change in direction
Final Answer: (4)

33) **Explaining :** This is a **conical pendulum**, where a mass rotates in a horizontal circle while the string makes an angle with the vertical. The string provides both the **centripetal force** (horizontal component of tension) and balances the **weight** (vertical component of tension). Using force components, we relate angular velocity to the **tension limit** of the string.
Concept : This question is based on conical pendulum:
Solution : Vertical force balance: $T\cos\theta = mg$
Horizontal (centripetal) force: $T\sin\theta = m\omega^2 r$
From triangle geometry:
$r = L\sin\theta$
So $T\sin\theta = m\omega^2(L\sin\theta)$
$T = mL\omega^2$

$$\omega = \sqrt{\frac{T}{mL}}$$

$$\omega = \sqrt{\frac{324}{0.5 \times 0.5}} = \sqrt{1296}$$

$\omega = 36$ rad/s
Final Answer : (4) 36 rad/s

34)

Explain Question : A stone tied to a 20cm string moves in a horizontal circle with centripetal acceleration 9.8 m/s^2 ; find its angular velocity.
Concept : This question is based on centripetal acceleration.
Solution :
$a_c = \omega^2 r$
Given $a_c = 9.8\ m/s^2$ and $r = 20\ cm = 0.2\ m$

$$\omega = \sqrt{\frac{a_c}{r}} = \sqrt{\frac{9.8}{0.2}} = \sqrt{49} = 7 rad/s.$$

Final answer : (1)

35) **Question Explanation:** This question explores the proportionality between gravity and the required angular velocity to maintain the friction-weight balance in a rotating system.
Concept: ω^2 is directly proportional to g. A change in g requires a square-root proportional change in ω.
Formula: $\omega = \sqrt{\frac{g}{\mu R}}$
Calculation: $\omega \propto \sqrt{g}$. If $g' = 0.9g$, then $\omega' = \sqrt{0.9}\omega \approx 0.948\omega$. This is a 5.1% decrease.

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Final Answer : Option (3)

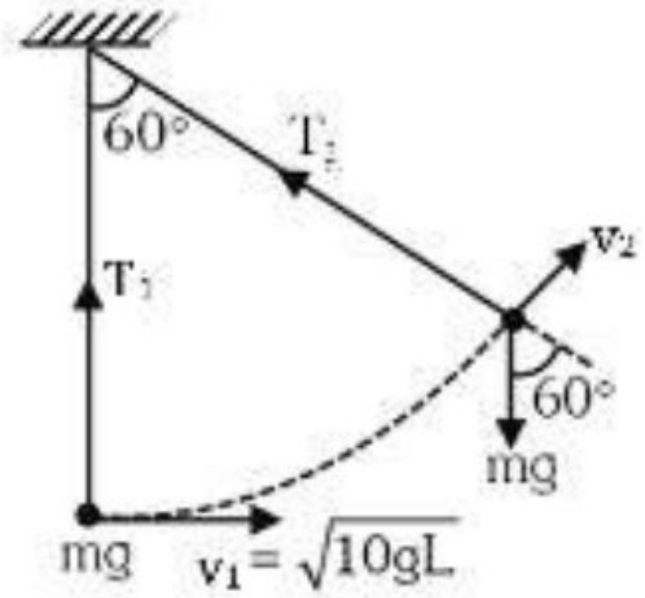
36) **Explain question:**

A ball (mass m) on string length L is given horizontal speed $\sqrt{10gL}$ at the lowest point. Find T_2/T_1 where T_1 is tension at bottom and T_2 is tension when string makes 60° with vertical.

Concept:

Use energy conservation to get speed at 60°. Radial (centripetal) equation: $T - mg\cos\theta = mv^2/L$.

Solution:



At bottom $v_1^2 = 10\ gL$.

$$T_1 = mg + \frac{mv_1^2}{L} = mg + \frac{m(10gL)}{L} = 11mg.$$

Height risen to 60°: $h = L(1 - \cos 60°) = \frac{1}{2} = L$

By energy: $v_2^2 = v_1^2 - 2gh = 10gL - 2g \cdot \frac{L}{2} = 9gL.$

Tension at 60°

$$T_2 = mg\cos 60° + \frac{mv_2^2}{L} = \frac{1}{2}mg + \frac{m(9gL)}{L} = \frac{19}{2}mg.$$

Ratio:

$$\frac{T_2}{T_1} = \frac{\frac{19}{2}mg}{11mg} = \frac{19}{22}.$$

Final Answer: Option (3)

37) **Explain the Question :** A stone moves in a vertical circle. Find the minimum speed at the top so that tension stays non-zero and the circle is completed.

Concept : This question is based on vertical circular motion.

Solution : For just completing the vertical circle, tension at the top is zero, so

$$\frac{mv^2}{r} = mg$$

$v^2 = gr$

Here $r = 5m$, $g = 9.8\ m/s^2$:

$v = \sqrt{9.8 \times 5} = \sqrt{49} = 7\ m/s$

Final Answer : (3)

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38) **Explain Question :**
A bob moves on a circular path; tension at top is zero. Find magnitude of acceleration at horizontal position A.

Concept :
This question is based on energy conservation, centripetal and tangential components of acceleration due to gravity.

Solution :
At top $v_{top}^2 = g\ell$ (since tension zero). Drop from top to A is height □, so by energy conservation.
$v_A^2 = v_{top}^2 + 2g\ell = g\ell + 2g\ell = 3g\ell$

Radial (centripetal) acceleration at A; $a_r = \frac{v_A^2}{\ell} = 3g.$ Tangential component : $a_t = g$. Resultant magnitude.
$a = \sqrt{a_r^2 + a_t^2} = \sqrt{(3g)^2 + g^2} = g\sqrt{10}$
with $g = 10\ m/s^2$, $a = 10\sqrt{10}\ m/s^2$.

Final Answer : Option (2)

39) **Explaining :** To complete the **vertical loop** without losing contact, the ball must have a **minimum speed at the top** of the circular part. At that point, the **normal force becomes zero**, and **centripetal force is provided entirely by weight**.
We use **energy conservation** to find the minimum height h from which the ball must be released so that it reaches that speed at the top.
Concept : This question is based on conservation of energy.
Solution:

- At top of loop, minimum speed condition:
$\frac{mv^2}{R} = mg \Rightarrow v_{top} = \sqrt{gR}$
- Use **energy conservation** from height h to top of loop (height 2R):
$mgh = \frac{1}{2}mv^2 + mg2R$

So, $mgh = \frac{1}{2}mv^2 + 2mgR$
Substitute $v^2 = gR$:
$mgh = \frac{1}{2}m(gR) + 2mgR$
$mgh = \left(\frac{1}{2}gR + 2gR\right)m = \frac{5}{2}mgR$
$\Rightarrow h = \frac{5}{2}R$
$h = \frac{5}{2} \times 0.1 = 0.25\ m$
Final Answer : (3) 0.25 m

40)

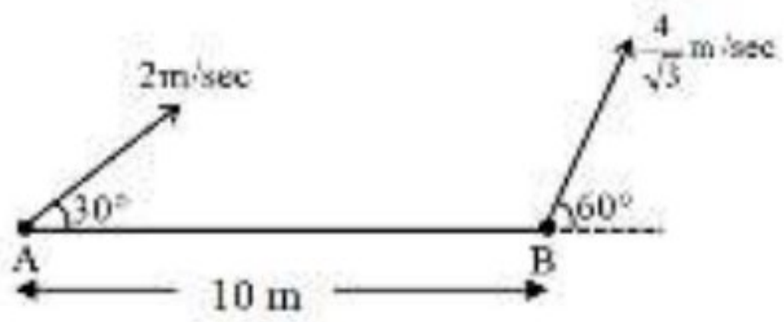
Question Explanation : Two particles A and B start from ground points 10 m apart. Their initial speeds and launch angles are given. Find the instantaneous angular speed of B about A

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at launch.

Concept : This question is based on relative angular speed

Solution :



relative velocity perpendicular to line joining

$$\omega_{rel} = \frac{}{\text{separation between them}}$$

$$\omega_{rel} = \frac{\frac{4}{\sqrt{3}}\sin 60^\circ - 2\sin 30^\circ}{10} = \frac{\frac{4}{\sqrt{3}}.\frac{\sqrt{3}}{2} - 2.\frac{1}{2}}{10}$$

$$\omega_{rel} = \frac{2-1}{10} = 0.1 \text{ rad/sec}$$

Final answer : Option (1)

41) **Explain the Question :** Find linear velocity using $\vec{v} = \vec{\omega} \times \vec{r}$.
Concept : This question is based on linear velocity in rotation is given by cross product.
Solution :

$\vec{\omega} = (3, -4, 1)$, $\vec{r} = (5, -6, 6)$

$$\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -4 & 1 \\ 5 & -6 & 6 \end{vmatrix} = -18\hat{i} - 13\hat{j} + 2\hat{k}$$

Final Answer : (2)

42)

Explain Question : How much the maximum safe speed increases if road radius becomes four times but banking angle stays same.

Concept : Banking of road
Formula : $v = \sqrt{Rg\tan\theta}$

Solution :

$$v \propto \sqrt{R}$$
$$\frac{v_2}{v_1} = \sqrt{\frac{4R}{R}} = 2$$

Percentage increase

$$= \frac{v_2 - v_1}{v_1} \times 100 = \left(\frac{v_2}{v_1} - 1\right) \times 100 = (2-1) \times 100 = 100\%$$

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Final answer : Option (4)

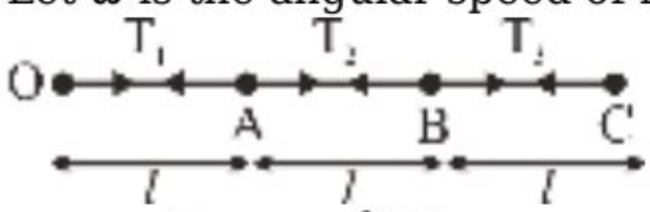
43)

Explain question: Find the ratio of tensions in three sections of a string connecting identical particles moving in a horizontal circle.

Concept: Tension provides centripetal force:
$T = m\omega^2 r$

Solution:

Let ω is the angular speed of revolution.



$T_3 = m\,\omega^3(3l)$

$T_2 - T_3 = m\,\omega^3\,2l \Rightarrow T_2 = m\omega^2\,(5l)$

$T_1 - T_2 = m\,\omega^2 l \Rightarrow T_1 = m\omega^2\,(6l)$

$T_3 : T_2 : T_1 = 3 : 5 : 6$

Final answer : Option (4)

44) **Explaining :** The centripetal force is provided by the tension in the string. The maximum speed is reached when the tension equals the breaking point of the string.

Concept : This question is based on From circular motion:

$$T = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{T \cdot r}{m}}$$

Solution : Given:

- T = 10 N
- m = 01 kg,
- r = 0.5 m

$$v = \sqrt{\frac{10 \times 0.5}{0.1}} = \sqrt{50}\ \text{m/s}$$

Final Answer : (2) $\sqrt{(50)}$ m/sec

45)

Question Explanation : A particle starts from rest at the top of a smooth sphere and slides down. We find the height where it loses contact.

Concept: Motion on a smooth sphere and condition for loss of contact.

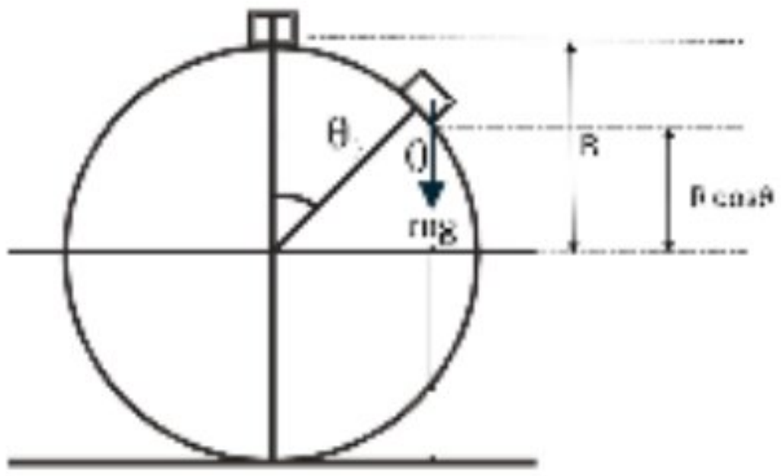
Solution:

The radius of the sphere be

$$R = \frac{30}{2} = 15m$$

at angle θ from the top, the particle leaves the surface when normal reaction becomes zero.

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Radial force balance.

$$mg\cos\theta = \frac{mv^2}{R} \quad \text{------- (i)}$$

Form energy conservation between top and angle θ:

$$mgR(1-\cos\theta) = \frac{1}{2}mv^2$$

$$2mg(1-\cos\theta) = \frac{mv^2}{R} \quad \text{------- (ii)}$$

From eq. (i) and (ii)

$mg\cos\theta = 2mg - 2mg\cos\theta$

$3\ mg\cos\theta = 2\ mg$

$$\cos\theta = \frac{2}{3}$$

Height from bottom:

$$h = R(1+\cos\theta) = 15\left(1+\frac{2}{3}\right) = 25m$$

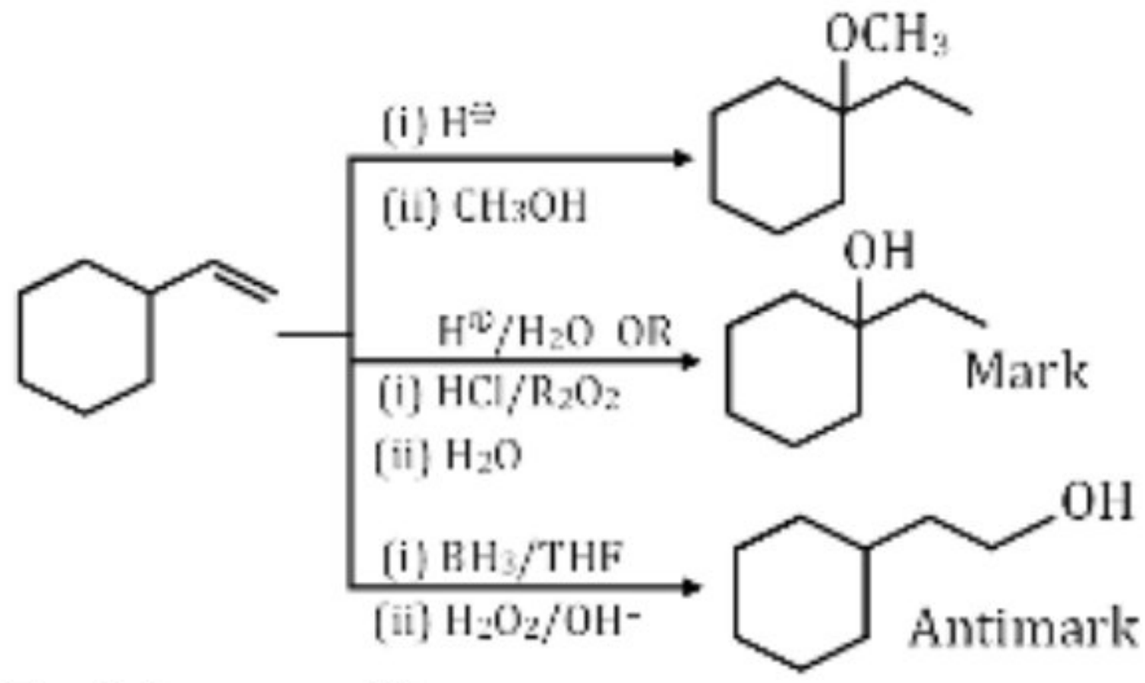
Final Answer: Option (1)

CHEMISTRY

46) **Explaining :** Match the column.

Concept : This question is based on prep. of Alcohol from Alkene.

Solution :



Final Answer : (1)

47)

Question Explanation: Lucas test results for different alcohols.

Concept: Reactivity order of alcohols.

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Solution:

The Lucas test converts alcohols to insoluble alkyl halides via an S_N1 mechanism; the rate depends on carbocation stability ($3^\circ > 2^\circ > 1^\circ$).

3° Alcohol $\xrightarrow[\text{(Lucas reagent)}]{\text{anhy } ZnCl_2+HCl}$ Immediate Turbidity

1° Alcohol $\xrightarrow[\text{Room Temperature}]{\text{anhy } ZnCl_2+HCl}$ No reaction

Statement I is true: Tertiary (3°) alcohols form the most stable 3º carbocations and react immediately (instant turbidity).

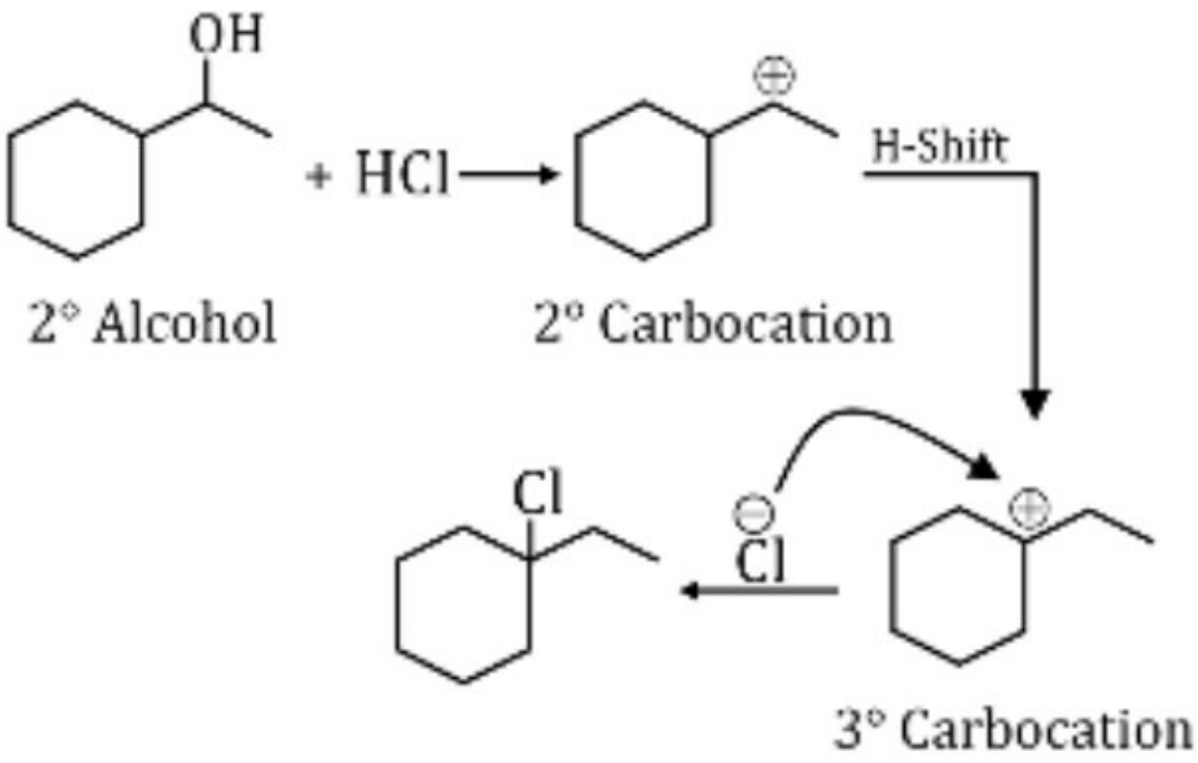
Statement II is true: Primary (1º) alcohols form the least stable 1º carbocations and require heating to react, so they do not produce turbidity at room temperature.

Final Answer: Option (1)

48) **Question Explanation:** The reaction involves the treatment of 1-cyclohexylethanol with hydrobromic acid (HBr) to determine the major organic product formed.

Concept: Nucleophilic substitution, Carbocation formation, 1,2-hydride shift, Rearrangement, Stability.

Solution:



Final Answer: Option (4)

49)

Question Explanation :

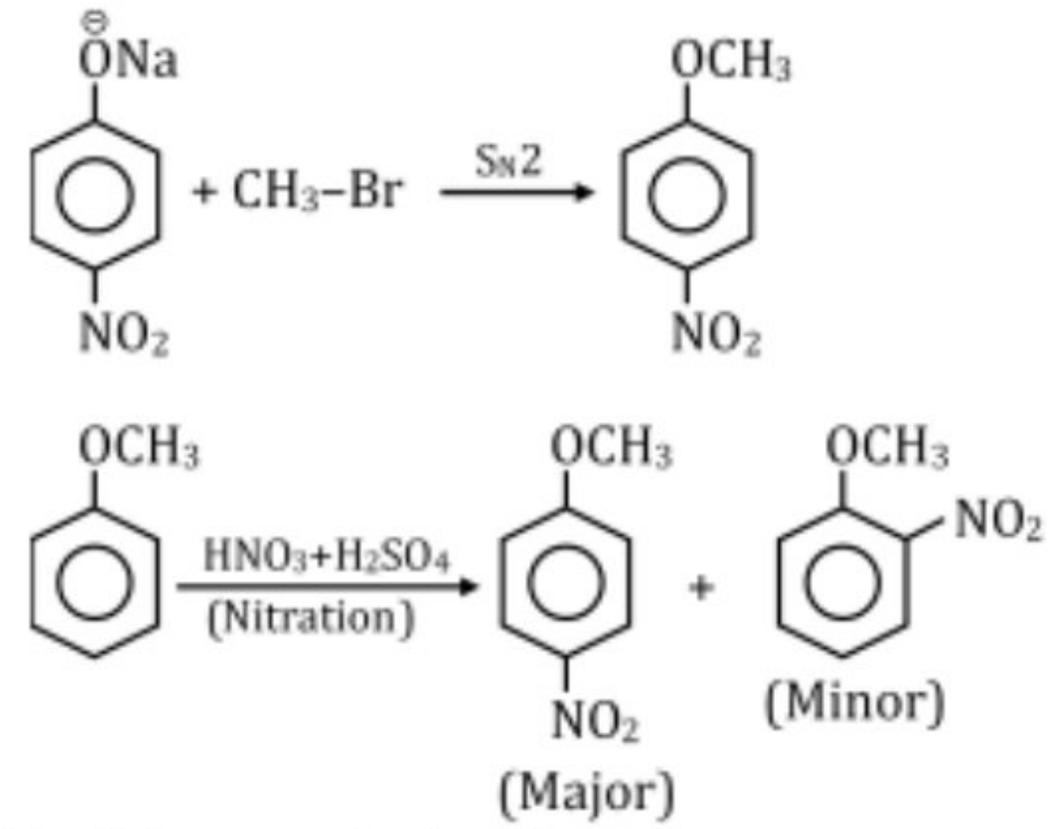
Which of the following is appropriate set of reactants for preparation of 1-Methoxy-4-nitrobenzene.

Concept :

Williamson ether synthesis, Nitration (ESR)

Solution :

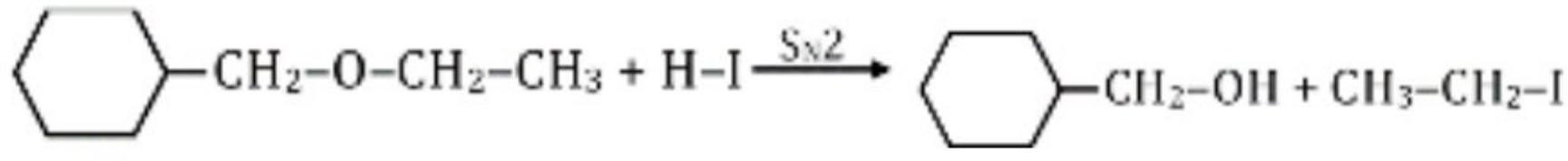
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Final Answer : Option(4)

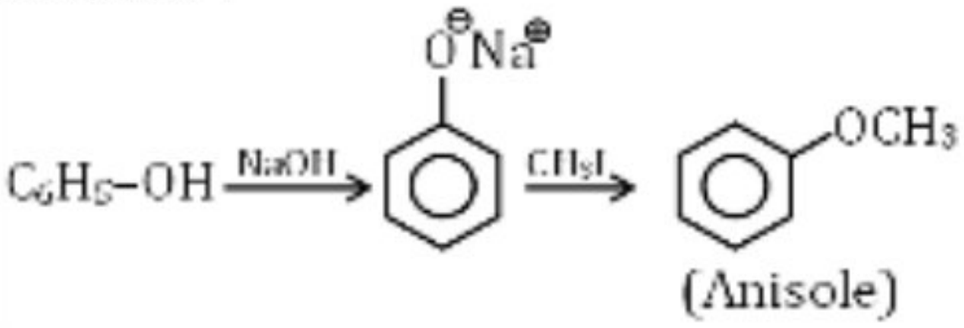
50)

Explaining : Find out final product
Concept : This question is based on Ether bond cleavage
Solution :



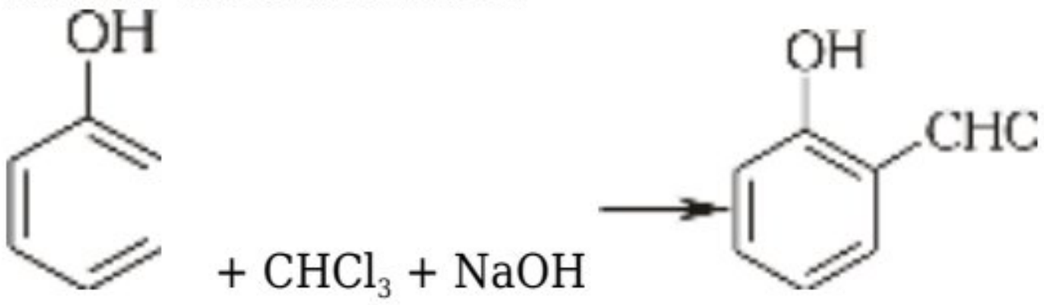
Final Answer : (3)

51) **Explaining :** Among the following sets of reactants which one produces anisole?
Concept : This question is based on preparation of ether.
Solution :



Final Answer : (2)

52) **Explaining :** Reaction of phenol with chloroform in presence of dilute sodium hydroxide finally introduces which one of the following functional group ?
Concept : This question is based on R.T. reaction.
Solution :
Reimer Tieman reaction

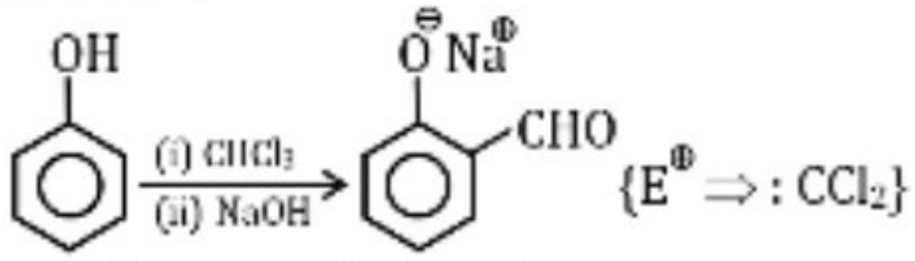


Final Answer : (2)

53)

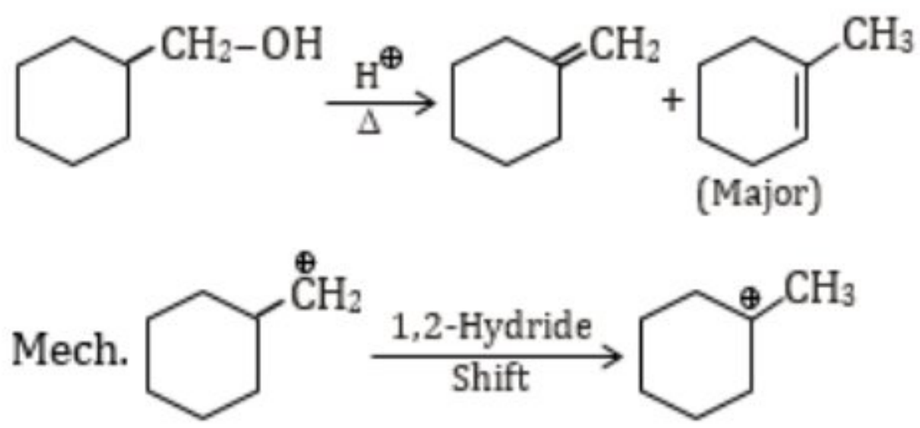
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Explain Question: In the reaction the electrophile involved is.
Concept: Reimer Tiemann reaction.
Solution:



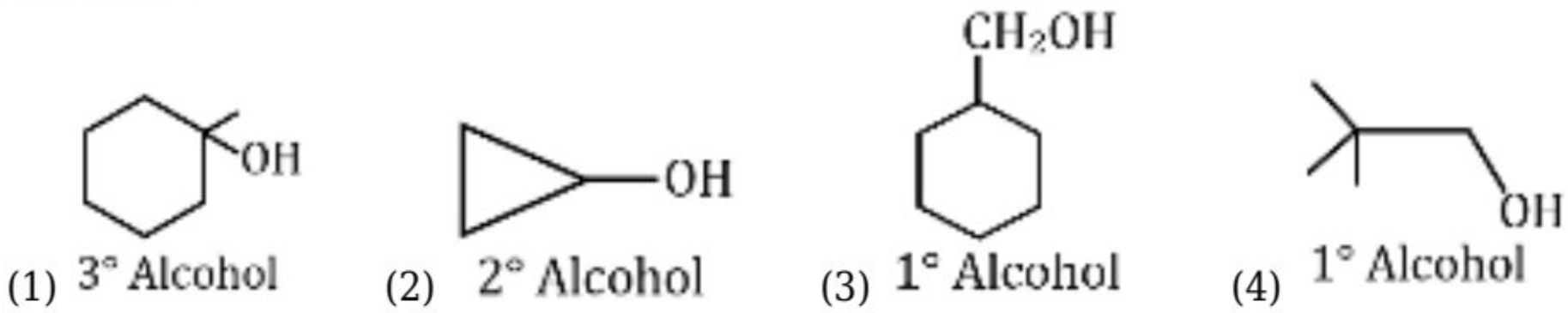
Final Answer: Option (4)

54) **Explaining :** Final product will be
Concept : This question is based on Elimination of reaction (dehydration of alcohol)
Solution :



Final Answer : (3)

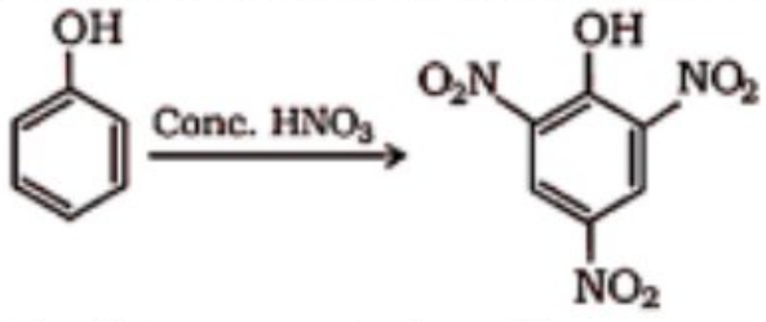
55) **Explaining :** Identify which of the following compound is 2° alcohol ?
Concept : This question is based on Type of alcohol.
Solution :



Final Answer : (2)

56)

Question Explanation: Determine the product formed when phenol reacts with concentrated nitric acid (conc. HNO_3).
Concept Used: Electrophilic Aromatic Substitution, Activating effect of hydroxyl (-OH) group, Nitration of Phenols.
Solution:
Concentrated HNO_3: Leads to explosive, multi-stage nitration due to high reactivity.



Final Answer: Option (4)

57)

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Question Explanation: Intermolecular dehydration product with two alcohols.

Concept: S_N2 intermolecular dehydration.

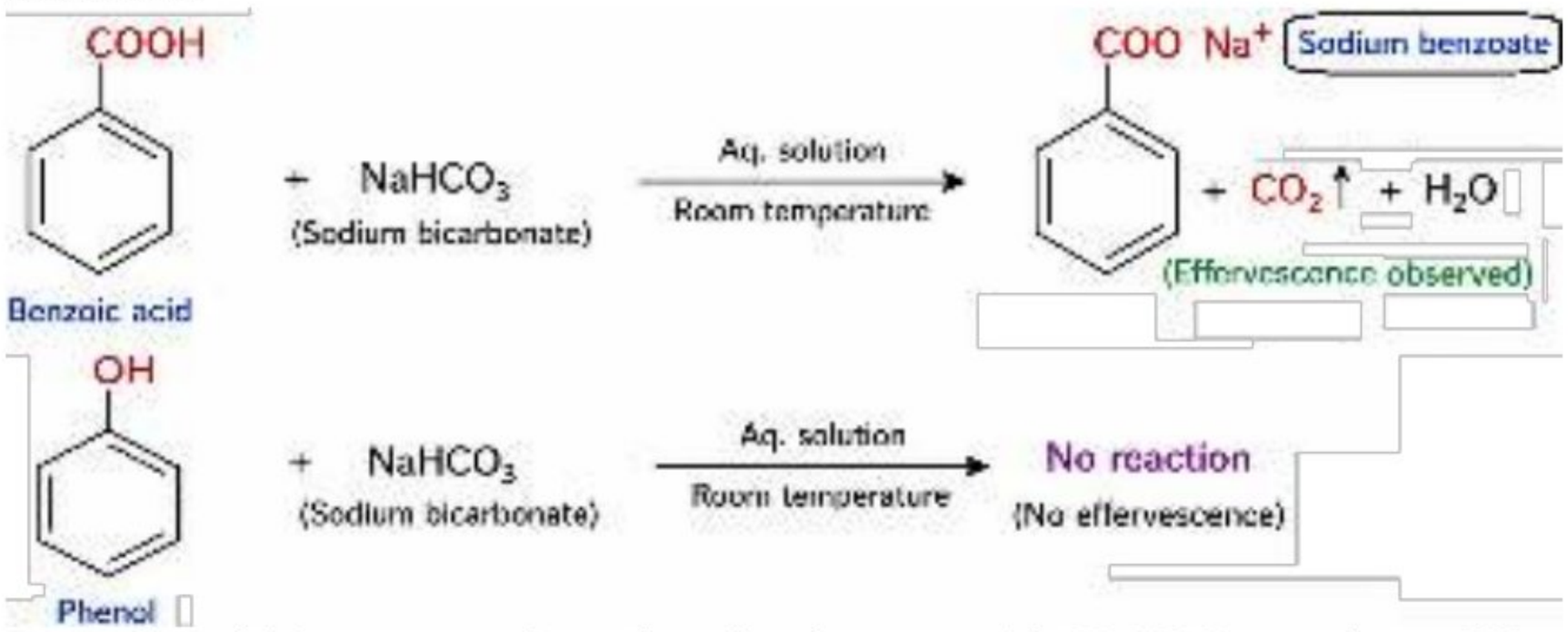
Solution:
The reaction of a mixture of two alcohols (Ethanol and Benzyl alcohol) with acid and heat (H^+/Δ) promotes intermolecular dehydration to form ethers. All three possible symmetrical and unsymmetrical ethers are formed:
1. $\text{Ethanol} + \text{Ethanol} \rightarrow \text{Diethyl ether } (CH_3CH_2OCH_2CH_3)$. (Option 1 - Obtained)
2. $\text{Benzyl alcohol} + \text{Benzyl alcohol} \rightarrow \text{Dibenzyl ether } (Ph-CH_2OCH_2-Ph)$. (Option 2 - Obtained)
3. $\text{Ethanol} + \text{Benzyl alcohol} \rightarrow \text{Ethyl benzyl ether } (Ph-CH_2OCH_2CH_3)$. (Option 3 - Obtained)

Option 4, $Ph-CH_2-O-CH_2-O-CH_3$, is a complex acetal/ether derivative (Benzyl (methoxymethyl) ether) that cannot be formed by the direct simple dehydration of the two given alcohols.

Final Answer: Option (4)

58)

Question Explanation: Identify the reagent that can distinguish **phenol** from **benzoic acid**.
Concept: Acidic Strength, Sodium Bicarbonate Test, Phenol, Carboxylic Acid
Solution:



Benzoic acid is stronger than phenol and reacts with **$NaHCO_3$** to release **CO_2 gas** with effervescence.
Phenol does not react with $NaHCO_3$ because it is too weak an acid.
Final Answer: Option (4)

59)

Question Explanation: Identify the name of the reaction in which **sodium phenoxide reacts with CO_2** to form **sodium salicylate**.

Concept: Kolbe–Schmitt Reaction, Carboxylation of Phenol

Solution: Sodium phenoxide reacts with **CO_2 at about 140°C under pressure** to form **sodium salicylate**, which on acidification gives **salicylic acid**. This carboxylation reaction is known as the **Kolbe–Schmitt reaction**.

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Final Answer: Option (2)

60)

Question Explanation:
Determine the Correct decreasing order of boiling points (B.P.) for the given set of organic compounds.

Concept:
Boiling Point, Intermolecular Forces, Molecular Weight.

Solution:
Boiling point is determined by the strength of the intermolecular forces (IMFs), All compounds have similar molecular weights (Propane = 44, Dimethyl ether = 46, Acetaldehyde = 44, Ethanol = 46), making the comparison purely based on IMFs.

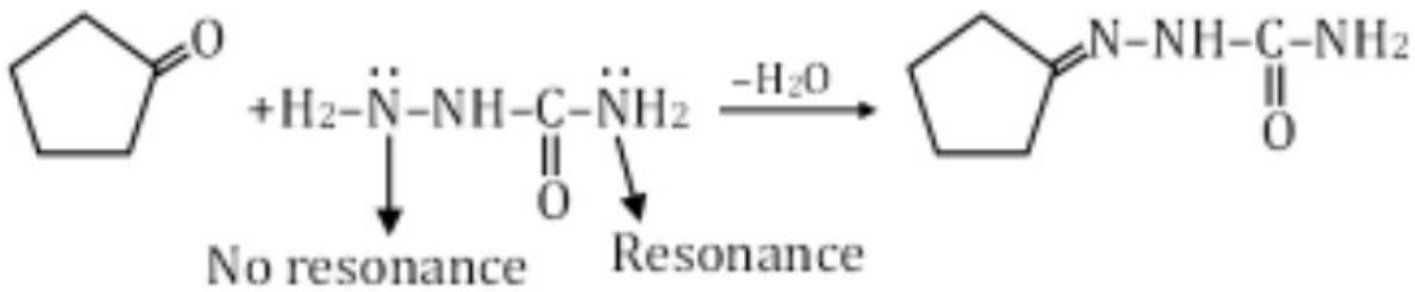
Compound	Molecular Formula	Primary Intermolecular Force	B.P. Ranking
Ethanol	CH_3CH_2OH	Hydrogen Bonding (Strongest)	**Highest**
Acetaldehyde	CH_3CHO	Dipole-Dipole Interaction (Strong)	Second Highest
Dimethyl ether	CH_3OCH_3	Dipole-Dipole Interaction (Weak/Moderate)	Third Highest
Propane	$CH_3CH_2CH_3$	London Dispersion Forces (Weakest)	**Lowest**

The Correct decreasing order is: $CH_3CH_2OH > CH_3CHO > CH_3OCH_3 > CH_3CH_2CH_3$

Final Answer: Option (4)

61)

Explaining : Identify product "P"
Concept : This question is based on Nucleophilic addition.
Solution :



Final Answer : (2)

62)

Explaining : Match the column.
Concept : This question is based on name reaction.
Solution : (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
Final Answer : (4)

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63)

Question Explanation:
The question asks to identify which carboxylic acid derivative is the most reactive towards Nucleophilic Acyl Substitution (S_N Ar or S_NAE), where a nucleophile attacks the carbonyl carbon and a leaving group is displaced.
Concept:
Acyl group, Nucleophilicity, Leaving group ability, Electrophilicity, Inductive effect, Resonance.

Solution:
The reactivity of acid derivatives depends on the electrophilicity of the carbonyl carbon and the leaving group ability of the substituent (L).

In an Acid halide (R – COCI), the Chlorine atom is highly electronegative (–I effect) and a very weak base, making it an excellent leaving group. This makes the carbonyl carbon highly electron-deficient.

The general order of reactivity is:
Acid Halide > Acid Anhydride > Ester > Amide

Chemical Reaction:
$R – COCI + Nu^- \rightarrow R – CONu + Cl^-$

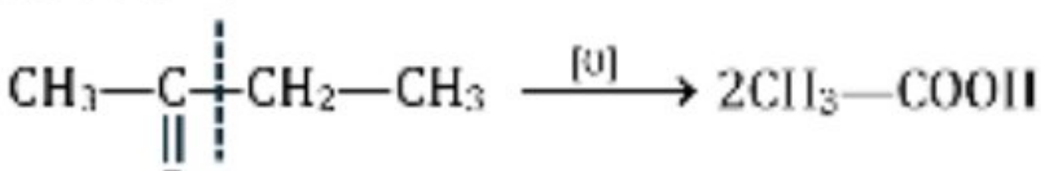
Final Answer: Option (1)

64)

Explaining : Brady's reagent is
Concept : This question is based on DNP test
Solution : An aq. Solution of 2,4-dinitrophenyl hydrazine (DNP) ⇒ Brady's reagent
Final Answer : (3)

65)

Explaining : Identify product (A)
Concept : This question is based on oxidation of ketone.
Solution :

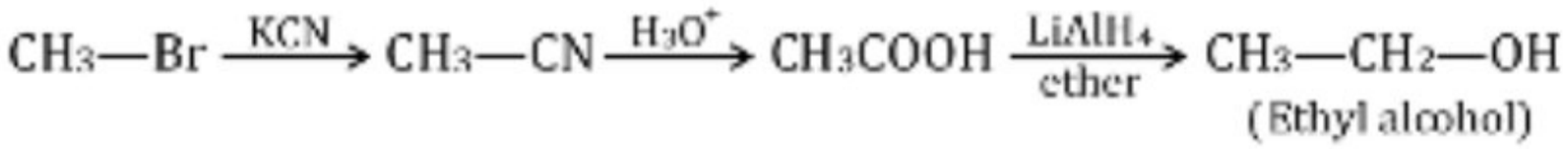


Final Answer : (1)

66)

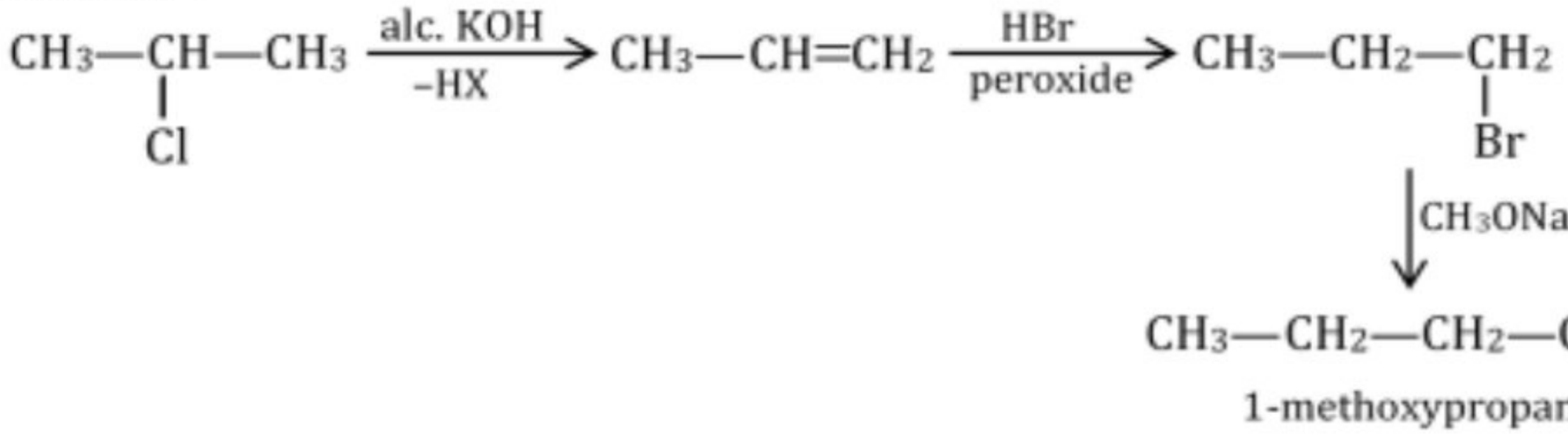
Explaining : Identify product (C)
Concept : This question is based on Reduction of carbonyl compound.
Solution :

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Final Answer : (2)

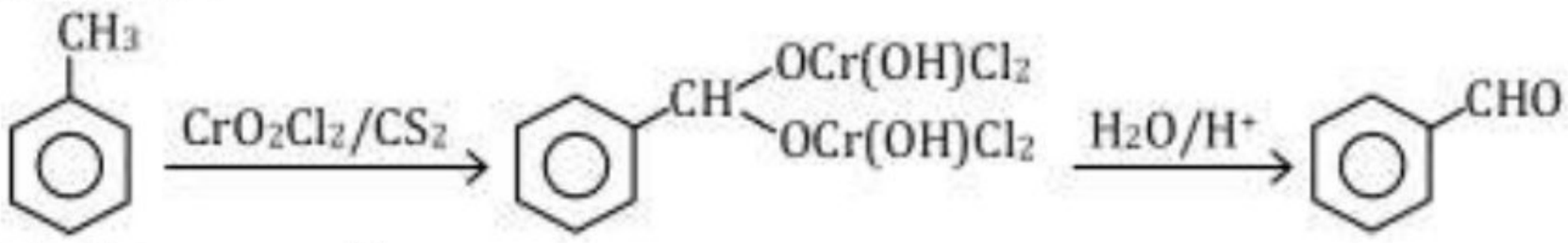
67) **Explaining :** Identify product (C).
Concept : This question is based on Nucleophilic substitution.
Solution :



Final Answer : (2)

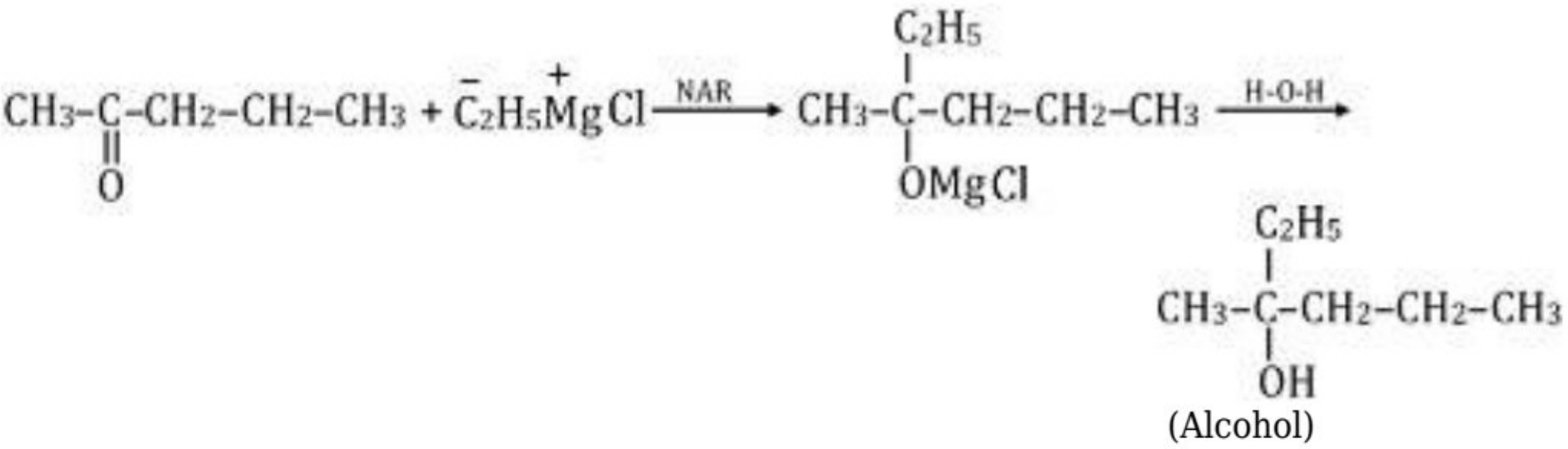
68)

Explaining : Identify product (B)
Concept : This question is based on preparation of aldehyde.
Solution :



Final Answer : (2)

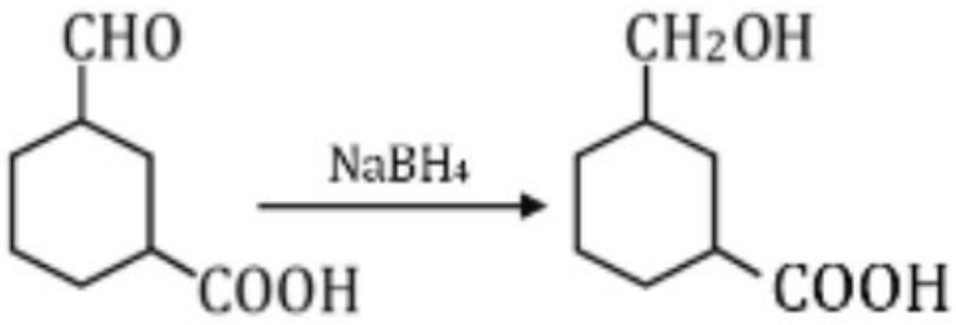
69) **Explaining :** Identify product.
Concept : This question is based on preparation of alcohol.
Solution :



Ketone + G.R. → 3° Alcohol
Final Answer : (4)

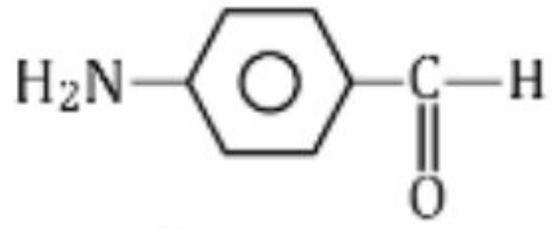
70) **Explaining :** Identify product.
Concept : This question is based on Reduction of aldehyde.
Solution :

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Final Answer : (1)

71) **Explaining :** Least reactive
Concept : This question is based on reactivity order of carbonyl compound.
Solution :



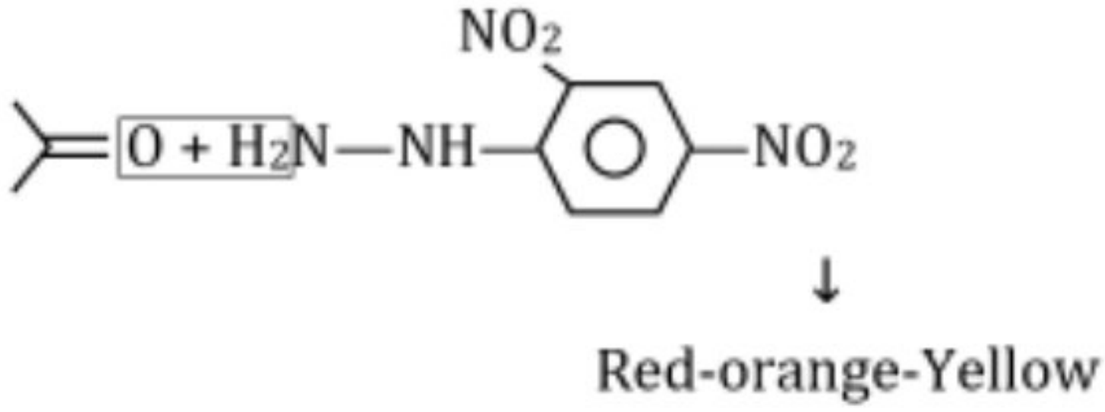
+M of $\ddot{N}H_2$ decreases electrophilicity of carbon.
Final Answer : (1)

72)

Explaining : Correct statement.
Concept : This question is based on mechanism of Cannizzaro Reaction.
Solution : The transfer of hydride ion to the carbonyl group
Final Answer : (2)

73)

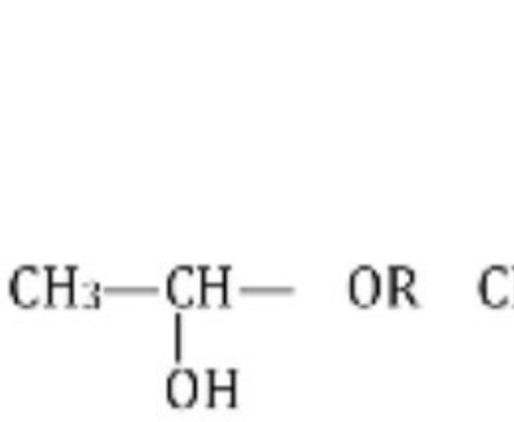
Explaining : Statement based question.
Concept : This question is based on Nucleophilic addition elimination reaction.
Solution :



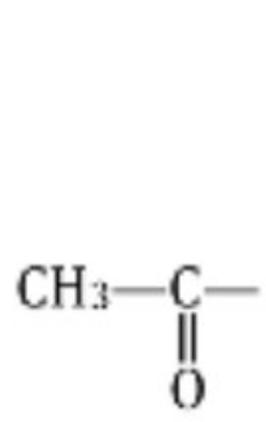
I[st] addition then elimination.
Final Answer : (3)

74) **Explaining :** Identify correct pair.
Concept : This question is based on iodoform test.
Solution :
Iodoform test given by

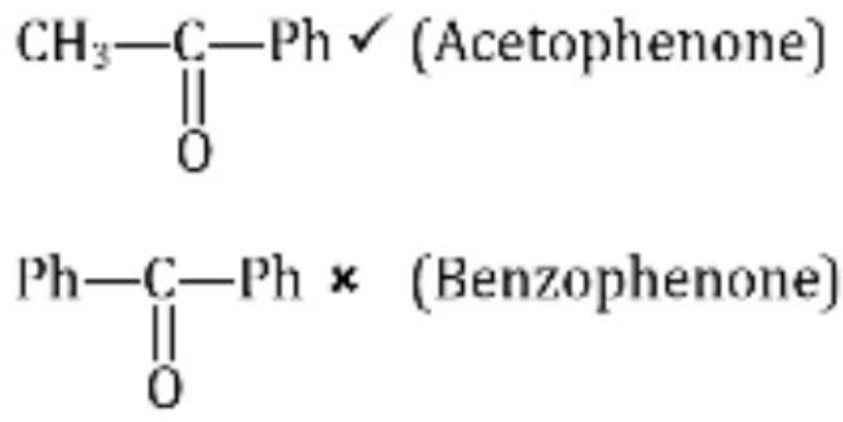
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OR



group



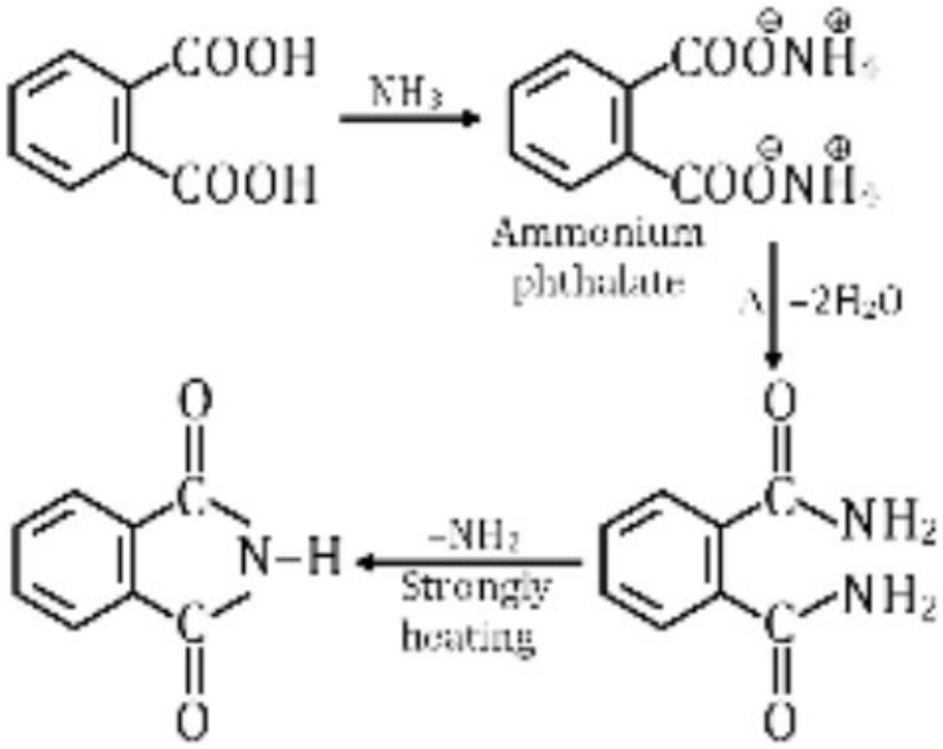
Final Answer : (2)

75)
Explaining : Identify product (A)
Concept : This question is based on aldol condensation.
Solution :



Final Answer : (2)

76) **Explaining :** Find product of given reaction.
Concept : This question is based on carboxylic acid reaction with NH_3 then strong heating.
Solution :



Final Answer : (2)

77) **Explaining :** Chronyl chloride test.
Concept : This question is based on Chronyl chloride test.
Solution : CrO_2Cl_2 and +6
Final Answer : (2)

78) **Explaining :** Color of ions
Concept : This question is based on Color of ions.
Solution : Unpaired e^- in Cu^{+2} Cr^{+3} Sc^+.
Final Answer : (4)

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79) **Explaining :** Properties of d & f block.
Concept : This question is based on Properties of d & f block.
Solution :
Zr ≈ Hf Lanthanide contraction.
Actinium → d-Block
Final Answer : (2)

80) **Explaining :** Magnetic nature.
Concept : This question is based on Magnetic nature.
Solution : Yb^{2+} → No unpaired e^-.
Final Answer : (2)

81) **Explaining :** Magnetic nature.
Concept : This question is based on Magnetic nature.
Solution : $\sqrt{n(n+2)}$ B.M.
Final Answer : (4)

82) **Explaining :** Properties of $KMnO_4$
Concept : This question is based on Properties of $KMnO_4$
Solution : Charge transfer when oxygen gives its electrons to Mn making it +VI
Final Answer : (3)

83) **Explaining :** Magnetic nature.
Concept : This question is based on Magnetic nature.
Solution : $\sqrt{n(n+2)}$ B.M.
Final Answer : (3)

84) **Explaining :** Electron configuration
Concept : This question is based on electron configuration.
Solution : Gd : [Xe] $4f^7$, $5d^1$, $6s^2$
Final Answer : (4)

85) **Explaining :** Properties of transition metals.
Concept : This question is based on Properties of transition metals.
Solution :
Cu, Ag, Au, Are coinage metals
Option (1) is correct. **Final Answer :** (1)

86)

Question Explanation: The magnetic moment of a transition metal ion is given. We must identify the metal ion based on the number of unpaired electrons.

Concept: Spin-only magnetic moment: $\mu = \sqrt{n(n+2)}$ B.M.
where n = number of unpaired electrons.

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Solution:
Given μ = 3.87 B.M.
Try values of n:
For n = 3 → μ = $= \sqrt{3\,(3+2)} = \sqrt{15} \approx 3.87$ B.M.
So, the ion has **3 unpaired electrons**.
Check options (electronic configuration):

A. $Cr^{2+} \rightarrow 3d^4$ (4 unpaired)

B. $Mn^{2+} \rightarrow 3d^5$ (5 unpaired)

C. $V^{2+} \rightarrow 3d^3$ (3 unpaired)

D. $Ti^{2+} \rightarrow 3d^2$ (2 unpaired)

Final Answer: Option (3)

87) **Explaining :** Oxidation states
Concept : This question is based on oxidation states.
Solution : Mn^{+2} to Mn^{+7}
Final Answer : (1)

88) **Explaining :** Chemical properties of 'd' block element.
Concept : This question is based on Chemical properties of 'd' block element.
Solution :

$$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} \underset{\text{green}}{Cr_2O_3} + N_2 + H_2O$$

Final Answer : (2)

89) **Explaining :** Lanthanide contraction.
Concept : This question is based on Lanthanide contraction.
Solution : Lanthanide contraction.
Final Answer : (1)

90) **Explaining :** Oxidation state
Concept : This question is based on oxidation state.
Solution :
Ce^{+4} = [Xe] $4f^0$
Tb^{+4} = [Xe] $4f^7$
Final Answer : (4)

BIOLOGY

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91) **Solution :** Statement I is correct (describes the fibrous root system in monocots like wheat). Statement II is incorrect because roots arising from the radicle form primary/tap roots; adventitious roots arise from parts of the plant other than the radicle (e.g., grass, *Monstera*, banyan tree).
Final Answer : (3)

92) **Solution :** Statement I is incorrect because leaflets attached at the tip of the petiole describe a *palmately* compound leaf (e.g., silk cotton); in pinnately compound leaves, leaflets are arranged on a common axis called the rachis (e.g., neem). Statement II is correct.
Final Answer : (4)

93) **Solution :** Statement I is correct. Statement II is incorrect because while most monocot seeds are endospermic, some (such as orchids) are non-endospermic
Final Answer : (3)

94) **Solution :** Growth of the main floral axis stops in cymose inflorescence precisely because the apex develops into/terminates in a flower. Flowers are borne in basipetal order
Final Answer : (1)

95) **Solution :** Ray florets of sunflower are epigynous. By definition, epigynous flowers have an inferior ovary because the thalamus encloses and fuses with the ovary, placing other floral parts above it.
Final Answer : (1)

96) **Solution :** Assertion is false because a staminode is a sterile stamen. Reason is true (e.g., epipetalous stamens in brinjal).
Final Answer : (1)

97) **Solution :** Alternate phyllotaxy is found in China rose, mustard, and sunflower. *Calotropis* and guava exhibit opposite phyllotaxy.
Final Answer : (1)

98) **Solution :**

A. Marginal → Pea
B. Axile → Lemon / Tomato / China rose
C. Parietal → Mustard / Argemone
D. Free central → Dianthus / Primrose
E. Basal → Sunflower / Marigold

Final Answer : (2)

99) **Solution :** Root hairs form as delicate thread-like structures from epidermal cells in the region of maturation.
Final Answer : (4)

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100) **Solution :** In some leguminous plants, the leaf base may become swollen, which is called the pulvinus.
Final Answer : (1)

101) **Solution :** The pattern/arrangement of veins and veinlets in the lamina is termed venation
Final Answer : (3)

102) **Explain Question:** The question asks that which of the given option is not the modification of the stem.
Concept: Modification of stem.
Solution: Pitcher of *Nepenthes* is not the modification of stem. (It is leaf modification) So, the correct answer is: 3.
Final Answer : (3)

103) **Solution :** In racemose inflorescences, the main axis continues to grow and flowers are borne laterally in acropetal succession.
Final Answer : (3)

104) **Solution :** Pea, gulmohur, bean, and *Cassia* have zygomorphic flowers. Mustard, datura, and chilli are actinomorphic.
Final Answer : (4)

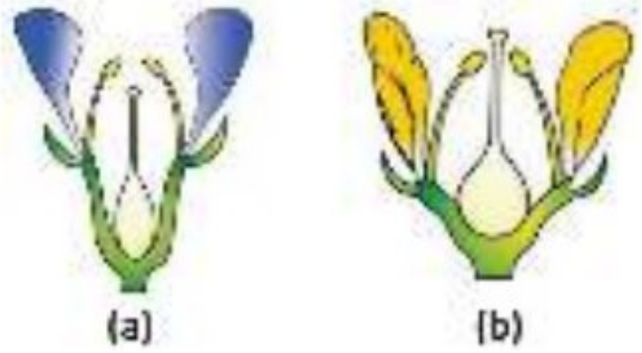
105)

The correct answer is: **3. Nodulated roots**
Explanation:

A. Nodulated roots are roots that have symbiotic nodules, where nitrogen-fixing bacteria, such as *Rhizobium*, are found. These bacteria help in fixing atmospheric nitrogen into a form that the plant can use for growth.

The other types of roots (fusiform, napiform, and conical} do not typically have nitrogen-fixing bacteria associated with them.

106) **Question Explanation:** The question asks you to identify the flower types based on the position of the calyx, corolla and androecium with respect to the ovary.



Concept : In a perigynous flower, the floral parts arise around the ovary at approximately the same level. In this condition, the ovary remains half-inferior/half-superior in appearance.

Solution :

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If gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level, it is called perigynous. The ovary here is said to be half inferior, e.g., plum, rose, peach.

Figure (a) → The floral parts are arranged around the ovary → Perigynous.
Figure (b) → The floral parts are also arranged around the ovary in the same manner → Perigynous.
Therefore, both figures represent perigynous flowers.
Final Answer : Option 4: (a)–Perigynous; (b)–Perigynous

107) **Solution :** The largest posterior petal is the standard (overlaps two lateral wings, which overlap two anterior keels).
Final Answer : (3)

108) **Solution :** When stamens are attached to the perianth (e.g., lily), they are termed epiphyllous.
Final Answer : (2)

109) **Solution :** Stamens united into more than two bundles are polyadelphous (e.g., *Citrus*)
Final Answer : (3)

110) **Solution :** Carpels are free (apocarpous) in lotus and rose. Mustard and tomato are syncarpous.
Final Answer : (3)

111) **EXPLANATION** If the floral diagram you are referring to belongs to the **Brassicaceae** family, then it will display specific characteristics typical of this family:
Key Features of Brassicaceae (Mustard family):
1. Symmetry: Actinomorphic (radial symmetry), meaning the flower can be divided into symmetrical halves in multiple ways.
2. Petals: There are **four petals** arranged in a **cross-like** (cruciform) pattern, which is a defining feature of this family.
3. Stamens: Typically, **six stamens** in total, often arranged in a **4+2** pattern, where four are longer and two are shorter.
4. Ovary: The ovary is usually **superior** and can develop into a capsule.
Given this, the family corresponding to the floral diagram you are referring to is indeed **Brassicaceae.**
Therefore, the correct answer to the question is:
2. Brassicaceae.

112) **Solution :** A fruit formed without fertilisation is called a parthenocarpic fruit.
Final Answer : (1)

113) **Solution :** Both mango and coconut are drupes, developing from monocarpellary syncarpous superior ovaries.
Final Answer : (3)

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114) **Question Explanation :** The question asks you to match each type of aestivation with its correct plant example.
Concept : Aestivation describes the arrangement of sepals or petals in a floral bud.
Solution:
(a) Imbricate → *Cassia* → (ii)
(b) Valvate → *Calotropis* → (i)
(c) Vexillary → Bean → (iv)
(d) Twisted → Cotton → (iii)
Therefore, the correct matching is (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii).
Final Answer: Option 2: (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

115) **Solution :** 'C' is radicle which forms root system upon germination.
Final Answer : (3)

116) **Solution :** In dicot roots, the vascular cambium is completely secondary in origin, arising from tissues located just below the phloem bundles and portions of pericycle tissue lying above the protoxylem.
Final Answer : (2)

117) **Solution :** Cells of medullary rays adjoining the intrafascicular cambium become meristematic and form the interfascicular cambium.
Final Answer : (1)

118)

Explain Question : The question asks about the main function of xylem vessels in plants.

Concept : Complex Permanent Tissue- Xylem

Explanation : The chief function of xylem vessels is to conduct sap, which primarily consists of water and dissolved minerals.

Final answer :Option (1)

119) **Solution :** Passage cells are unthickened endodermal cells situated opposite to protoxylem vessels to facilitate the movement of water into the vascular cylinder.
Final Answer : (2)

120) **Solution :** The correct order from exterior to interior in a dicot stem is: Epidermis → Hypodermis → General Cortex → Endodermis →Pericycle → Vascular bundles → Pith.
Final Answer : (2)

121) **Solution :** Casparian strips are composed of suberin deposited along the radial and tangential walls of endodermal cells, preventing apoplastic movement of water.
Final Answer : (3)

122) **Solution :** In conjoint vascular bundles, phloem is located only on the outer side of

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xylem.
Final Answer : (2)

123) **Solution :** Absence of cambium between xylem and phloem prevents secondary tissue formation, defining them as closed bundles
Final Answer : (2)

124)

Question Explanation: What are the sugar transport elements in gymnosperms and pteridophytes?

Concept: Phloem

Solution:

Gymnosperms and pteridophytes transport sugars through sieve cells, which are elongated phloem cells with tapering ends and pores, but lack companion cells. These are considered a more primitive type of conducting element compared to sieve tubes in angiosperms.

Final answer: 1

125) **Solution :** When bulliform cells lose water, they become flaccid, causing the leaf blades to curl inward to minimize exposed surface area and reduce water loss.
Final Answer : (2)

126) **Solution :** Monocot stems feature scattered conjoint closed vascular bundles, each enclosed by a sclerenchymatous bundle sheath.
Final Answer : (2)

127)

Option (A) is correct. It indicates dicot root



128) **Solution :** Exarch arrangement means protoxylem faces the periphery while metaxylem faces the center, which is characteristic of roots.
Final Answer : (2)

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129) **Solution :** In monocot stem vascular bundles, disintegration of inner protoxylem vessels leads to the formation of a water-containing cavity.
Final Answer : (3)

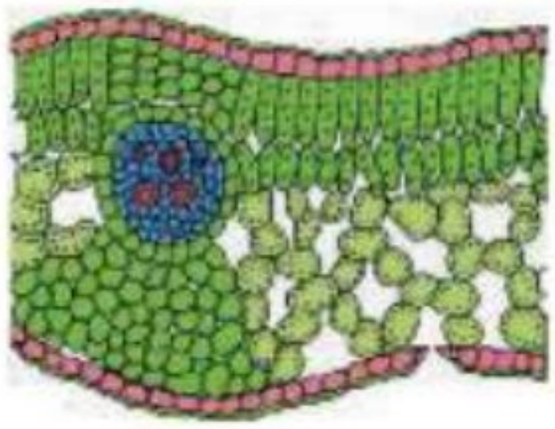
130) **Solution :** Isobilateral leaves (monocots) generally have stomata on both surfaces and an undifferentiated mesophyll layer.
Final Answer : (1)

131) **Solution :** Statement I is correct. Statement II is incorrect because vascular bundle size in dicot leaves depends on vein thickness, varying along the reticulate network.
Final Answer : (2)

132) **Solution :** Statement I is correct. Statement II is incorrect because monocot roots do not undergo secondary growth at all.
Final Answer : (1)

133) **Solution :** Statement I is false (radial bundles occur in roots, while conjoint bundles occur in stems and leaves). Statement II is correct.
Final Answer : (3)

134) **Question Explanation :** The question asks you to identify the type of leaf shown in the given diagram.



Concept : A hypostomatic leaf has stomata present only on the lower (abaxial) surface of the leaf.

Solution:
The given diagram represents a hypostomatic leaf, in which stomata are restricted to the lower surface.
Therefore, the correct identification is hypostomatic leaf.
Final Answer Option 1: Hypostomatic leaf

135) **Solution :** Assertion is correct. Reason is false because all vascular bundles in a monocot stem are enclosed by sclerenchymatous bundle sheaths.
Final Answer : (3)

136) **Solution :** Suberized Casparian strips block the apoplastic pathway, forcing solution through the selectively permeable cell membranes into the symplast
Final Answer : (1)

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137) **Solution :** Assertion is a direct anatomical fact. Reason is false because monocot leaves have prominent parallel veins and functional companion cells.
Final Answer : (3)

138) **Solution :** Polyarch xylem, large pith, and lack of secondary growth distinguish monocot roots from dicot roots. Casparian strips are present in the endodermis of both dicot and monocot roots.
Final Answer : (1)

139) **Solution :** All four listed characteristics are features of primary dicot stem anatomy
Final Answer : (3)

140) **Solution :** Ground tissue includes cortex, pericycle, pith, medullary rays, and mesophyll. Epidermis belongs to the epidermal tissue system.
Final Answer : (1)

141) **Solution :** Though potato and brinjal are different species, they share many common features, so they are grouped under the same genus *Solanum*.
Final Answer : (3)

142) **Solution :** Plant families are characterised using both vegetative (roots, stems, leaves) and reproductive (flowers, fruits) features.
Final Answer : (2)

143) **Solution :** Lower taxonomic categories share more common characteristics. As we move higher, similarities decrease, making classification more complex.
Final Answer : (3)

144) **Solution :** These animals share basic structural features like a notochord and a dorsal hollow neural system, which places them under phylum Chordata.
Final Answer : (3)

145) **Solution :** Species is the lowest category. Closely related species form a genus, related genera form a family, and related families form an order.
Final Answer : (2)

146) **Solution :** The order *Carnivora* includes families such as *Felidae* and *Canidae* that share a few common characters.
Final Answer : (4)

147) **Solution :** Binomial nomenclature consists of two components, genus and specific epithet, which together form the scientific name.
Final Answer : (1)

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148) **Solution :** Taxonomy includes several processes such as identification and classification, not just naming, so the reason is false.
Final Answer : (3)

149) **Solution :** ICBN is used for plants, ICZN for animals, binomial nomenclature has two words, and *Mangifera* is the genus name of mango.
Final Answer : (3)

150) **Solution :** Taxonomy includes classification and naming, systematics studies evolutionary relationships,taxa indicate different levels, and species is the lowest category.
Final Answer : (1)

151) **Solution :** NCERT emphasises morphological differences for identifying species. Reproductive isolation is not mentioned here and is not the only criterion.
Final Answer : (2)

152) **Solution :** The author’s name indicates the scientist who first described the species scientifically, not who discovered or named it casually.
Final Answer : (3)

153) **Solution :** Economic use was the basis of early classifications, not taxonomy. Taxonomy includes identification, classification, and nomenclature.
Sure! Here’s the **numbered Assertion–Reason MCQs starting from 38**, based on the paragraph you provided:
Final Answer : (4)

154) **Solution :** Systematics studies different kinds of organisms and their relationships, which includes evolutionary relationships, making the reason a correct explanation.
Final Answer : (1)

155) **Solution :** The term systematics originates from the Latin word “systema”, which means systematic arrangement, directly explaining the assertion.
Final Answer : (1)

156) **Solution :**
A – Mollusca: Triploblastic, coelomate, no notochord
B – Echinodermata: Coelomate, **but no notochord**
C – Cnidaria: Diploblastic, no coelom, notochord absent
D – Platyhelminthes: Triploblastic but **acoelomate**
Final Answer : (3)

157) **Solution :** Cockroach: belong to largest phylum, **open circulation**, **Malpighian tubules (uricotelic)**; no nephridia or respiratory pigment.
Final Answer : (1)

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158) **Solution :** All statements accurately contrast Annelida and Arthropoda features.
Final Answer : (2)

159) **Solution :** Water enters through **ostia**, moves into **spongocoel**, and exits through **osculum**.
Final Answer : (2)

160) **Solution:** Porifera lack a nervous system.
Platyhelminthes have a simple ladder-like nervous system.
Annelida have a ventral nerve cord with paired ganglia.
Arthropoda have a more complex nervous system with brain and sensory organs.
Final Answer : (1)

161) **Solution :**
1. *Physalia* → correct
2. *Ascaris* → **pseudocoelomate**, not acoelomate
3. Earthworm → correct
4. *Starfish* → correct
5. *Tapeworms* → are **hermaphroditic**, no trochophore larva
Final Answer : (2)

162) **Solution :** Echinoderm – adult radial symmetry, endoskeleton of ossicles, water vascular system and tube feet.
Final Answer : (2)

163) **Solution :** Metagenesis refers to alternation of generation between asexual and sexual phases of an organism.
Final Answer : (3)

164) **Solution :** Statement 1 is misleading as "always" is too strong — some lower vertebrates may have remnants of notochord.
Statement 4 is false — cephalochordates are considered more advanced.
Final Answer : (2)

165) **Solution :** A Sponges use choanocytes for feeding and water flow; they are asymmetrical.
B Cnidarians are diploblastic with cnidocytes.
C Sponges lack a nervous system and contractile tissues.
D Cnidarians have gastrovascular cavity and alternate polyp/medusa forms.
Final Answer : (1)

166) **Solution :**
Earthworm (Annelida): true coelom, nephridia, metamerism
Ascaris (Nematoda): pseudocoelomate, unsegmented, renette cells.
Final Answer : (1)

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167) **Solution :** Nematodes are pseudocoelomates, bilaterally symmetrical, unsegmented body and their excretion occurs via specialized renette cells.
Final Answer : (3)

168) **Solution :** Radula → feeding organ in molluscs
Proboscis → feeding and burrowing in *Balanoglossus* (Hemichordata)
Final Answer : (1)

169) **Solution :** In Urochordata, notochord is present only in the larval tail. *Ascidia*, *salpa*, *Doliolum*, *Herdmania* belongs to Urochordata.
Final Answer : (3)

170) **Solution :** Radula is a feeding organ in molluscs like snail.
Choanocytes are specialized cells in sponges used for filter feeding.
Final Answer : (2)

171) **Solution :** The shown organism is a jawless vertebrate-*Petromyzon* that has paired fins absent & has cartilagenous skeleton.
Final Answer : (2)

172) **Solution :** In Osteichthyes, four pairs of gills are present which are covered by an opercutum.
Final Answer : (2)

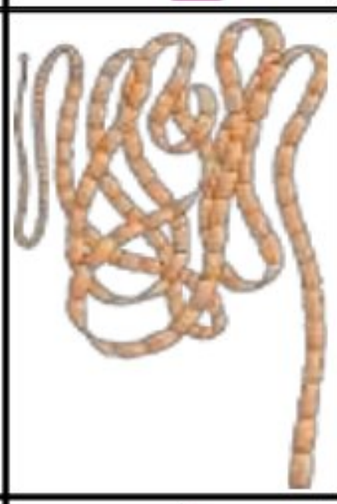
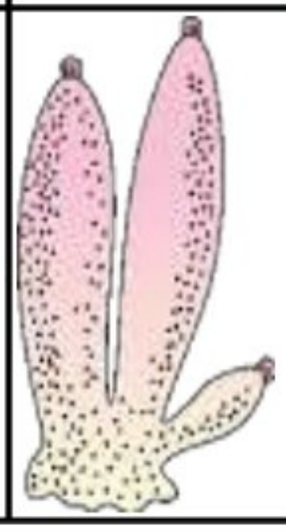
173) **Solution:**
Sycon → Porifera → cellular level
Hydra → Cnidaria → tissue level
Pleurobrachia → Ctenophora → tissue level
Ctenoplana → Ctenophora → tissue level
Therefore, **Sycon** is the odd one out.
Final Answer : (1)

174) **Solution:** Hydra belongs to Cnidaria.
Final Answer : (2)

175) **Solution: Cephalochordata:** Notochord extends from head to tail and persists throughout life.
Example → **Branchiostoma (Amphioxus)**
Vertebrata: Notochord occurs during embryonic development but is replaced by the vertebral column in adults.
Final Answer : (2)

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176) **Solution :** A-ii-b, B-i-a, C-iv-c, D-iii-d
Choose the correct match from column-I, II and III :-

Column-I		**Column-II**		**Column-III**	
A		i	Aschelminthes	b	Organ system level of organisation
B		iv	Coelenterata	a	Tissue level of organisation
C		i	Platyhelminthes	c	Organ/organ system level of organisation
D		iii	Porifera	d	Cellular level of organisation

Final Answer : (2)

177) **Solution :** Bilaterally symmetrical, triploblastic and coelomate animals.
Final Answer : (2)

178) **Solution :** A - *Ascaris*, B - Tapeworm, C - *Sycon*, D - *Pila*
Final Answer : (2)

179) **Solution :** Eyes have Eye lids
cloaca is present
Final Answer : (4)

180) **Solution :** Cyclostomes (e.g., lampreys and hagfish) have 6-15 pairs of gill slits for respiration. (1) is *Petromyzon* that belongs to class cyclostomata.
Final Answer : (1)

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